Introduction to Data Science and Engineering
- Data visualization in practice

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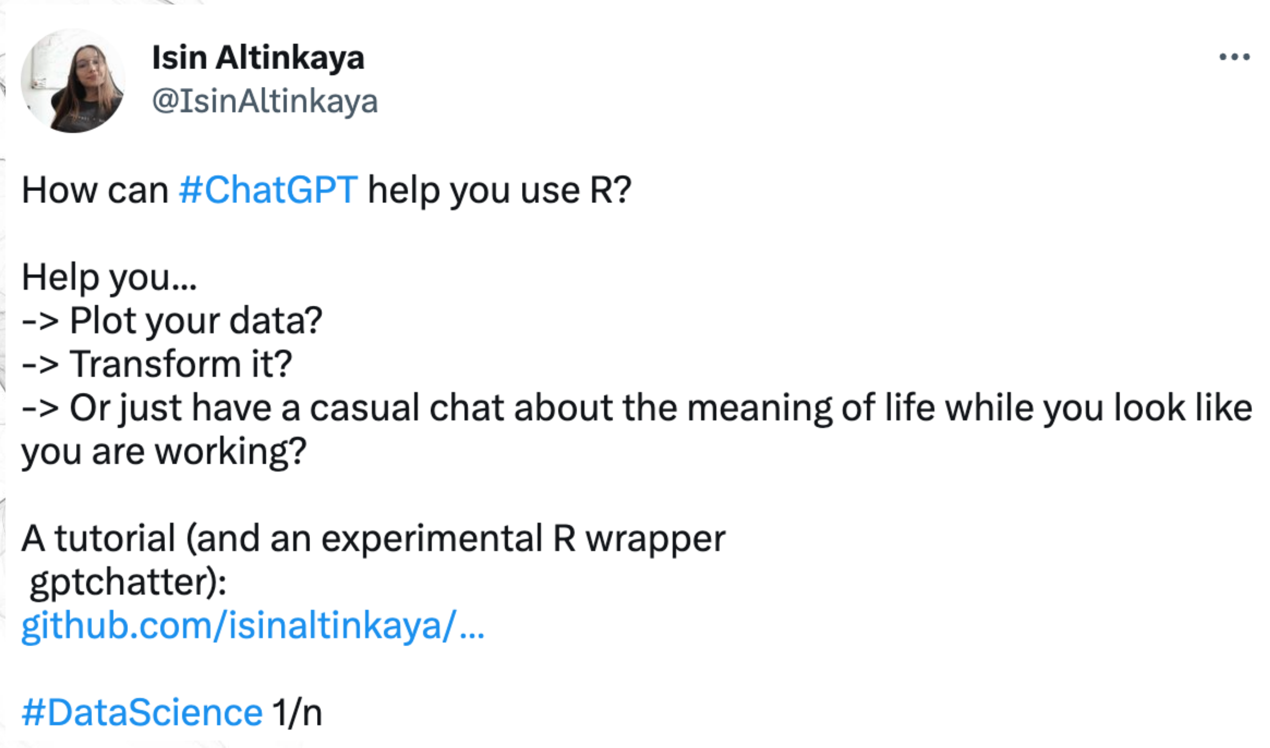
Some slides retrofitted by Michael Wu

2023’s slide: Shall I teach to write prompt instead of code?

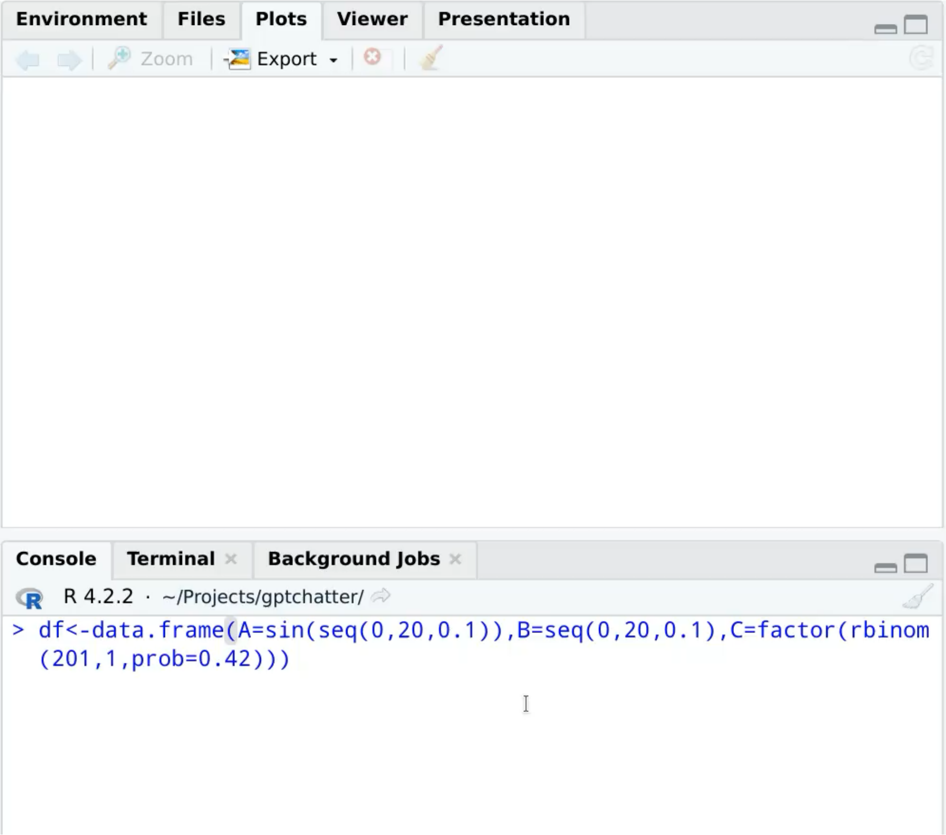

Isin Altinkaya
@IsinAltinkaya

How can #ChatGPT help you use R?

Help you...
-> Plot your data?
-> Transform it?
-> Or just have a casual chat about the meaning of life while you look like you are working?

A tutorial (and an experimental R wrapper
gptchatter):
github.com/isinaltinkaya/...

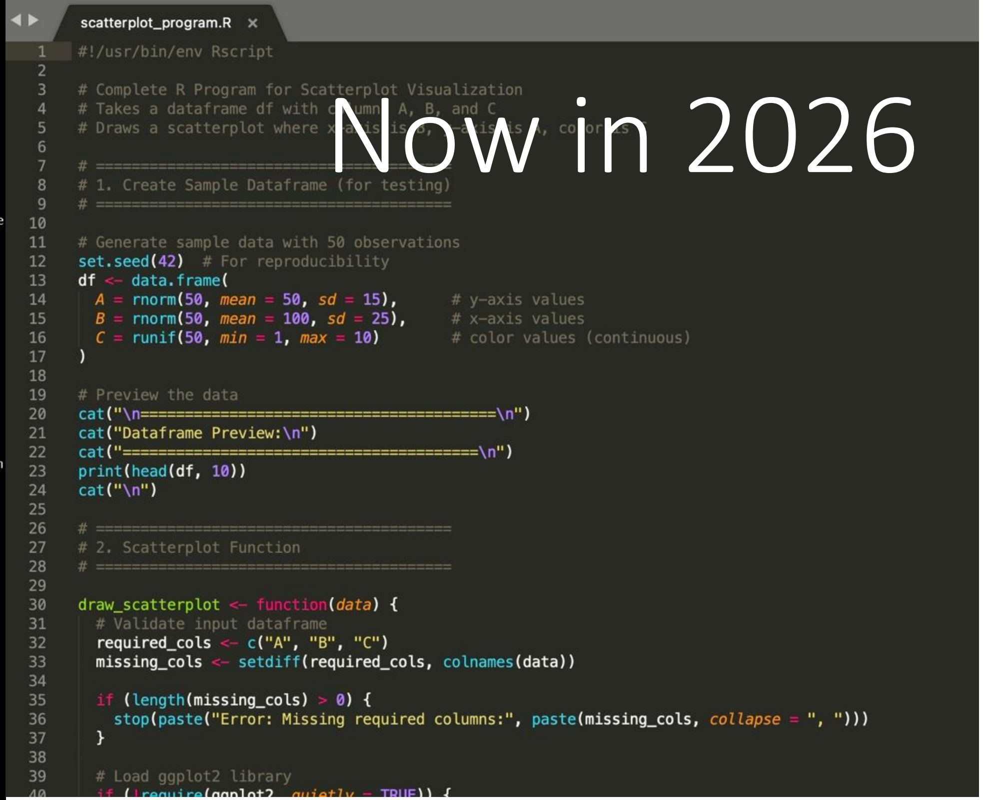
#DataScience 1/n

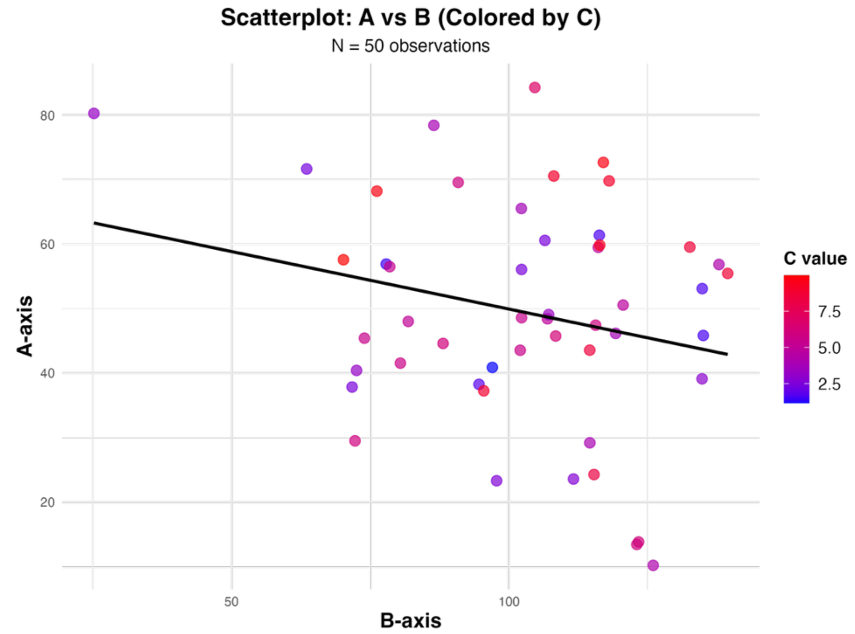


https://twitter.com/i/status/1599440535529623552

Now in 2026


No recent activity
mimo-v2-flash · API Usage Billing
~/Temp/ccTests
/model to try Opus 4.5
Write a complete program using R language. The program takes in a dataframe named df. df has 3 columns A, B, and C. Draw a scatterplot where x axis is B, y axis is A, color is C.
● I'll write a complete R program that creates a scatterplot with the specified requirements. Let me first explore what files exist in the current directory to understand the codebase structure.
● Search(pattern: "*")
⎿ Found 100 files (ctrl+o to expand)
● Let me check the current directory more clearly:
● Bash(ls -la /Users/rbluo/Temp/ccTests)
⎿ total 88
drwxr-xr-x 12 rbluo staff 384B Jan 29 11:17 .
drwxr-xr-x 26 rbluo staff 832B Jan 8 13:22 ..
… +10 lines (ctrl+o to expand)
● This is a test directory with shell scripts and markdown files. I'll write a complete R program that creates a scatterplot with the specified requirements. Let me create an R script file:
● Write(scatterplot_program.R)
⎿ Wrote 106 lines to scatterplot_program.R
#!/usr/bin/env Rscript
Complete R Program for Scatterplot Visualization
Takes a dataframe df with columns A, B, and C
Draws a scatterplot where x-axis is B, y-axis is A, color is C
====================================
1. Create Sample Dataframe (for testing)
====================================
… +96 lines (ctrl+o to expand)
● I've written a complete R program for creating a scatterplot. Let me run it to verify it works:
● Bash(Rscript scatterplot_program.R)
⎿ ==
Dataframe Preview:
==
… +30 lines (ctrl+o to expand)
● The program runs successfully. Let me verify the image was created:
● Bash(ls -la /Users/rbluo/Temp/ccTests/*.png 2>/dev/null || echo "No PNG files found")
⎿ -rw-r--r-- 1 rbluo staff 144K Jan 29 11:28 /Users/rbluo/Temp/ccTests/scatterplot_A_vs_B_by_C.png
● The R program has been successfully created and executed. The scatterplot has been generated and saved as scatterplot_A_vs_B_by_C.png.
Summary
I've written a complete R program (/Users/rbluo/Temp/ccTests/scatterplot_program.R) that:
1. Creates sample data - A dataframe df with 50 observations containing columns A, B, and C
2. Validates input - Checks that required columns exist
3. Draws a scatterplot using ggplot2:
- X-axis: Column B
- Y-axis: Column A
- Color gradient: Column C (blue to red)
- Adds linear trend line for insight
4. Displays and saves the plot as PNG file
The program is designed to work with any dataframe that has columns named A, B, and C. You can use it by passing your own dataframe:
# Example usage:
df <- data.frame(A = c(10, 20, 30), B = c(50, 60, 70), C = c(1, 5, 9))
source("scatterplot_program.R")
The output file is saved at /Users/rbluo/Temp/ccTests/scatterplot_A_vs_B_by_C.png
* Cogitated for 11m 59s
›
►► bypass permissions on (shift+tab to cycle)


scatterplot_program.R
#!/usr/bin/env Rscript
Complete R Program for Scatterplot Visualization
Takes a dataframe df with columns A, B, and C
Draws a scatterplot where x-axis is B, y-axis is A, color is C
==
1. Create Sample Dataframe (for testing)
==
Generate sample data with 50 observations
set.seed(42) # For reproducibility
df <- data.frame(
A = rnorm(50, mean = 50, sd = 15), # y-axis values
B = rnorm(50, mean = 100, sd = 25), # x-axis values
C = runif(50, min = 1, max = 10) # color values (continuous)
)
# Preview the data
cat("\n==\n")
cat("Dataframe Preview:\n")
cat("==\n")
print(head(df, 10))
cat("\n")
# ==
# 2. Scatterplot Function
# ==
draw_scatterplot <- function(data) {
# Validate input dataframe
required_cols <- c("A", "B", "C")
missing_cols <- setdiff(required_cols, colnames(data))
if (length(missing_cols) > 0) {
stop(paste("Error: Missing required columns:", paste(missing_cols, collapse = ", ")))
}
# Load ggplot2 library


Scatterplot: A vs B (Colored by C)
N = 50 observations
A-axis
B-axis
80
60
40
20
50
100
C value
7.5
5.0
2.5

Case study: new insights on poverty

Hans Rosling was the co-founder of the Gapminder Foundation, an organization dedicated to educating the public by using data to dispel common myths about the so-called developing world. The organization uses data to show how actual trends in health and economics contradict the narratives that emanate from sensationalist media coverage of catastrophes, tragedies, and other unfortunate events. As stated in the Gapminder Foundation's website:

Journalists and lobbyists tell dramatic stories. That's their job. They tell stories about extraordinary events and unusual people. The piles of dramatic stories pile up in peoples' minds into an over-dramatic worldview and strong negative stress feelings: "The world is getting worse!", "It's we vs. them!", "Other people are strange!", "The population just keeps growing!" and "Nobody cares!"

Hans Rosling
Professor of International Health
Karolinska Institute

https://en.wikipedia.org/wiki/Hans_Rosling

http://www.gapminder.org/

Gapminder's bubble chart

https://www.gapminder.org/tools/#$chart-type=bubbles&url=v2

Case study: new insights on poverty

Hans Rosling conveyed actual data-based trends in a dramatic way of his own, using effective data visualization. Following slides are based on two talks that exemplify this approach to education**: New Insights on Poverty and The Best Stats You've Ever Seen**. Specifically, in this section, we use data to attempt to answer the following question:

- **Is it a fair characterization of today's world to say it is divided into western rich nations and the developing world in Africa, Asia, and Latin America?**

https://www.ted.com/talks/hans_rosling_reveals_new_insights_on_poverty

https://www.ted.com/talks/hans_rosling_shows_the_best_stats_you_ve_ever_seen

Using the gapminder dataset

```
R 4.2.2 · ~/
> library(tidyverse)
> library(dslabs)
> data(gapminder)
> str(gapminder)
'data.frame':	10545 obs. of  9 variables:
 $ country         : Factor w/ 185 levels "Albania","Algeria",..: 1 2 3 4 5 6 7 8 9 10 ...
 $ year            : int  1960 1960 1960 1960 1960 1960 1960 1960 1960 1960 ...
 $ infant_mortality: num  115.4 148.2 208 NA 59.9 ...
 $ life_expectancy : num  62.9 47.5 36 63 65.4 ...
 $ fertility       : num  6.19 7.65 7.32 4.43 3.11 4.55 4.82 3.45 2.7 5.57 ...
 $ population      : num  1636054 11124892 5270844 54681 20619075 ...
 $ gdp             : num  NA 1.38e+10 NA NA 1.08e+11 ...
 $ continent       : Factor w/ 5 levels "Africa","Americas",..: 4 1 1 2 2 3 2 5 4 3 ...
 $ region          : Factor w/ 22 levels "Australia and New Zealand",..: 19 11 10 2 15 21
```

Hans Rosling's quiz

As done in the New Insights on Poverty video, we start by testing our knowledge regarding differences in child mortality across different countries. For each of the six pairs of countries below, which country do you think had the highest child mortality rates in 2015? Which pairs do you think are most similar?

1. Sri Lanka or Turkey
2. Poland or South Korea
3. Malaysia or Russia
4. Pakistan or Vietnam
5. Thailand or South Africa

Let's go find out

```
R 4.2.2 · ~/
> gapminder |>
+ filter(year == 2015 & country %in% c("Sri Lanka","Turkey")) |>
+ select(country, infant_mortality)
    country infant_mortality
1 Sri Lanka              8.4
2    Turkey             11.6
```

Turkey has the higher infant mortality rate

Do it to all countries, we got

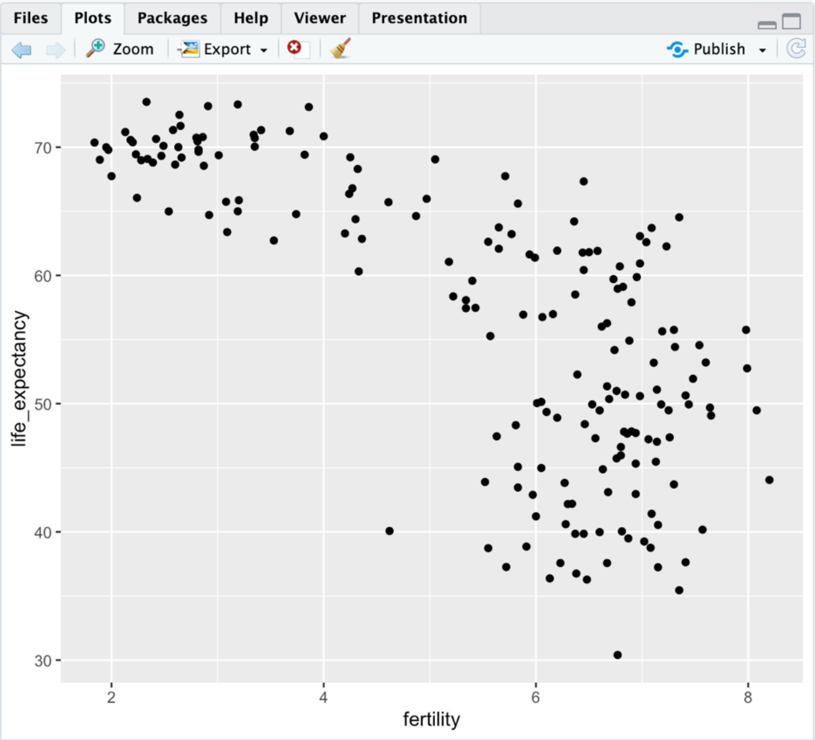
country	infant mortality	country	infant mortality
Sri Lanka	8.4	Turkey	11.6
Poland	4.5	South Korea	2.9
Malaysia	6.0	Russia	8.2
Pakistan	65.8	Vietnam	17.3
Thailand	10.5	South Africa	33.6

It turns out that when Hans Rosling gave this quiz to educated groups of people, **the average score was less than 2.5 out of 5**, worse than what they would have obtained had they guessed randomly. **This implies that more than ignorant, we are misinformed**

Scatterplots

The reason for this stems from the preconceived notion that the world is divided into two groups: the western world (Western Europe and North America), characterized by long life spans and small families, versus the developing world (Africa, Asia, and Latin America) characterized by short life spans and large families. But do the data support this dichotomous view?

In order to analyze this world view, our first plot is a scatterplot of life expectancy versus fertility rates (average number of children per woman). We start by looking at data from 60 years ago, when perhaps this view was first cemented in our minds



R 4.2.2 · ~/

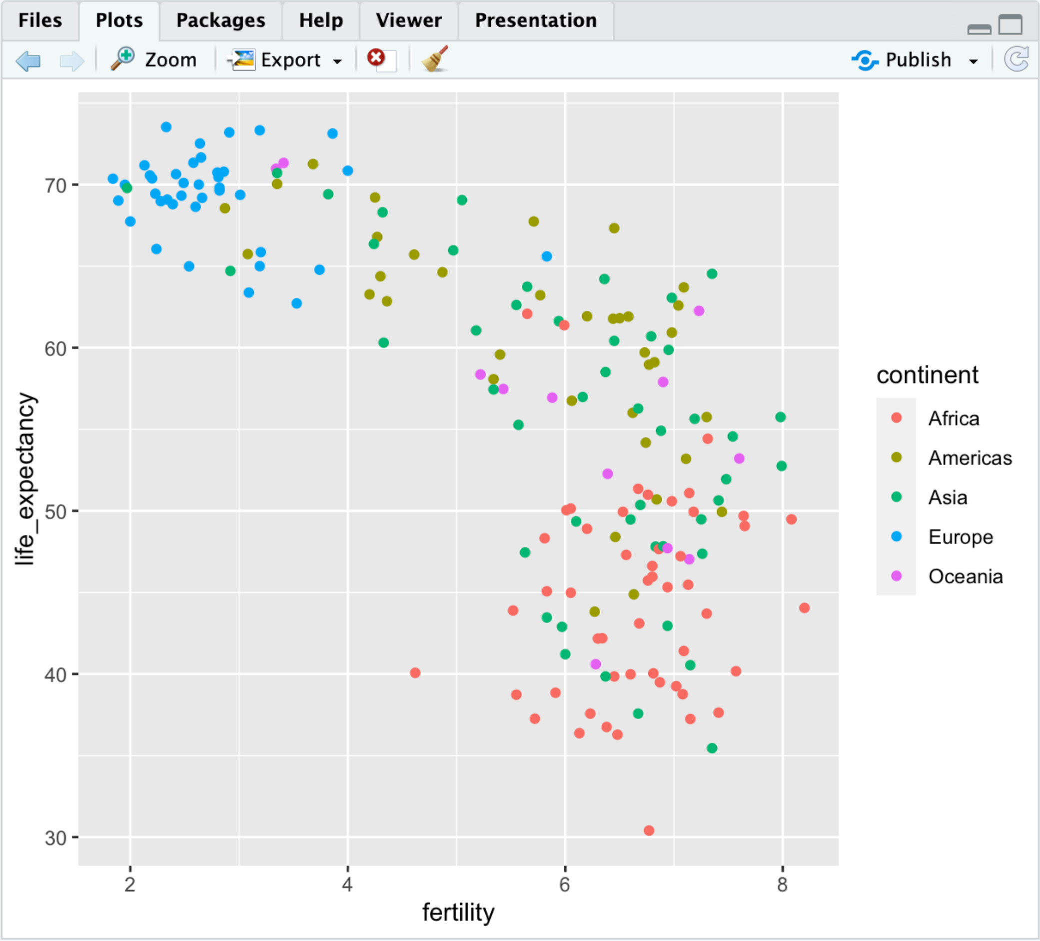
```
> filter(gapminder, year == 1963) |>
+ ggplot(aes(fertility, life_expectancy)) +
+ geom_point()
```

Most points fall into two distinct categories:

- Life expectancy around 70 years and 3 or fewer children per family
- Life expectancy lower than 65 years and more than 5 children per family

Scatterplots

To confirm that indeed these countries are from the regions we expect, we can use color to represent continent



R 4.2.2 · ~/

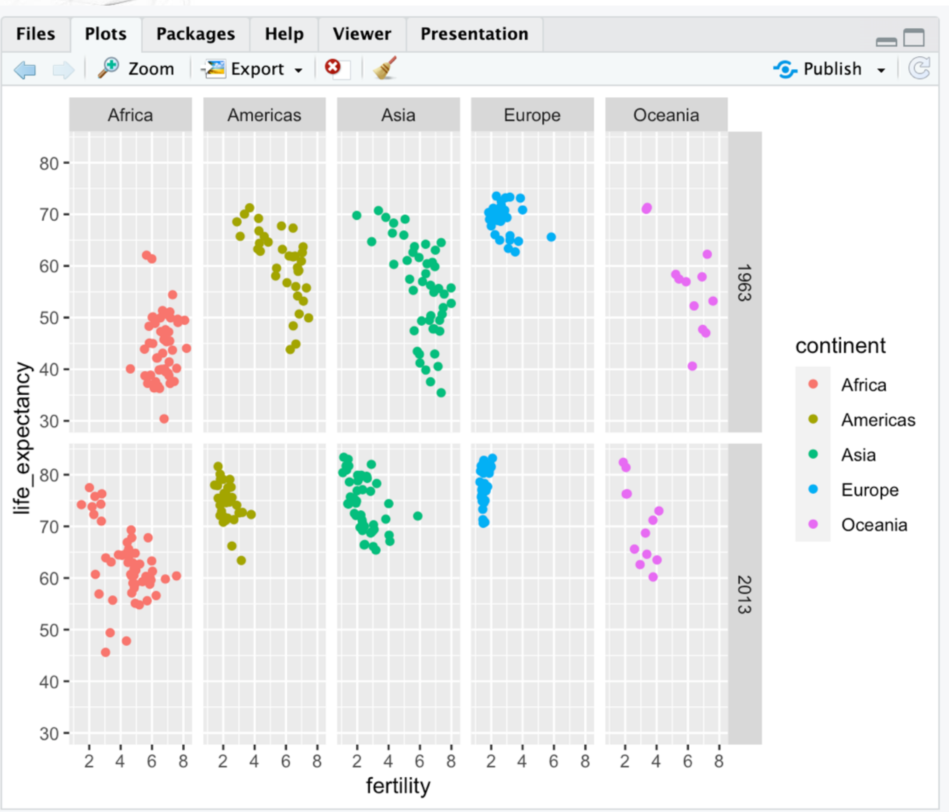
```
> filter(gapminder, year == 1963) |>
+ ggplot( aes(fertility, life_expectancy, color = continent)) +
+ geom_point()
```

In 1963, “the West versus developing world” view was grounded in some reality. Is this still the case 50 years later?

Faceting

We could easily plot the 2012 data in the same way we did for 1962. To make comparisons, however, side by side plots are preferable. In ggplot2, we can achieve this by faceting variables: we stratify the data by some variable and make the same plot for each strata

To achieve faceting, we add a layer with the function facet_grid, which automatically separates the plots. The function expects the row and column variables to be separated by a '~'



R 4.2.2 · ~/

```
> filter(gapminder, year %in% c(1963, 2013)) |>
+ ggplot(aes(fertility, life_expectancy, color = continent)) +
+ geom_point() +
+ facet_grid(year~continent)
```

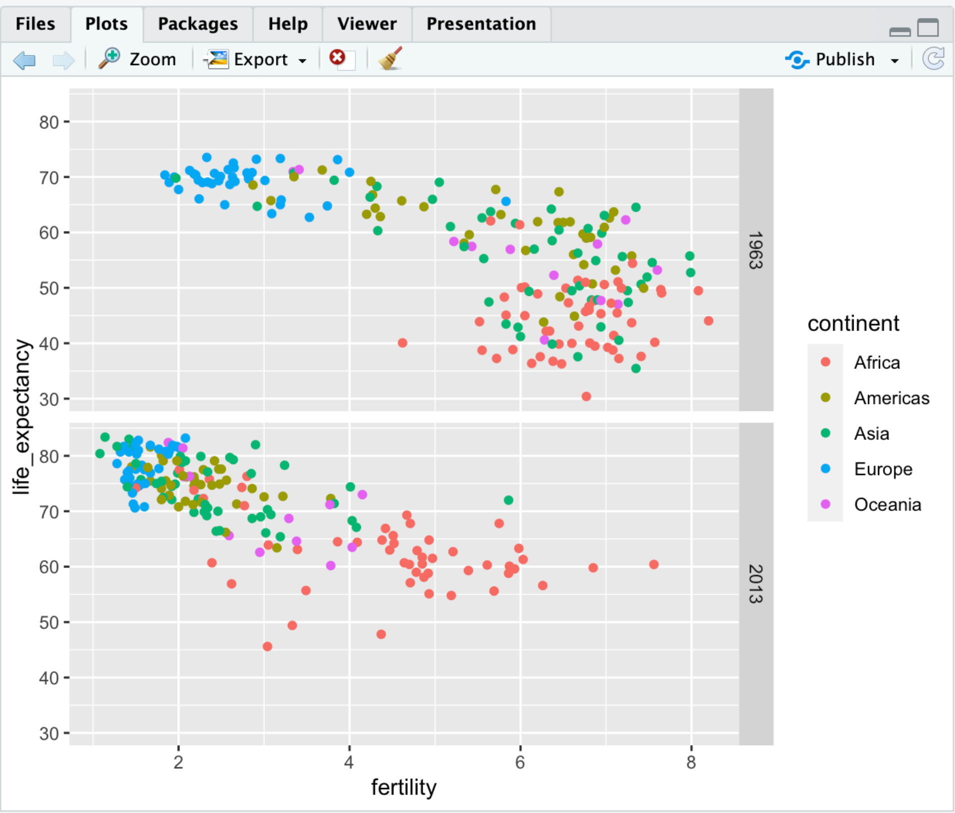
We see a plot for each continent/year pair. However, this is just an example and more than what we want, which is simply to compare 1962 and 2012

Faceting

In this case, there is just one variable and we use . to let facet know that we are not using one of the variables

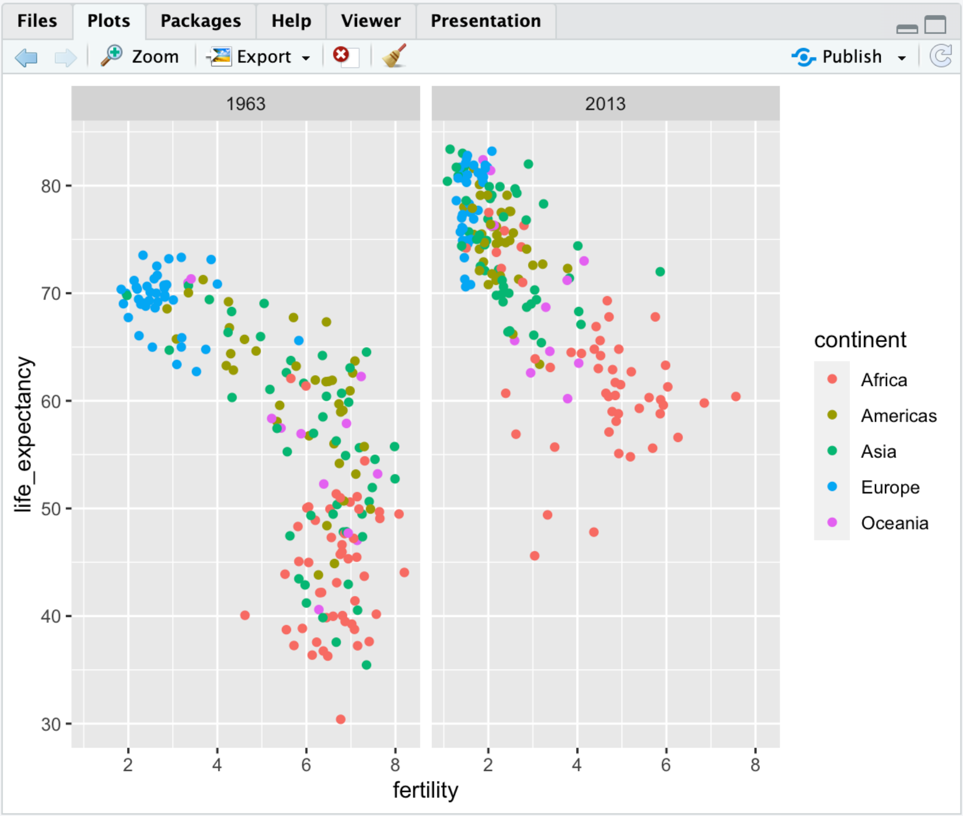
R 4.2.2 · ~/

```
> filter(gapminder, year%in%c(1963, 2013)) |>
+ ggplot(aes(fertility, life_expectancy, col = continent)) +
+ geom_point() +
+ facet_grid(year ~ .)
```



R 4.2.2 · ~/

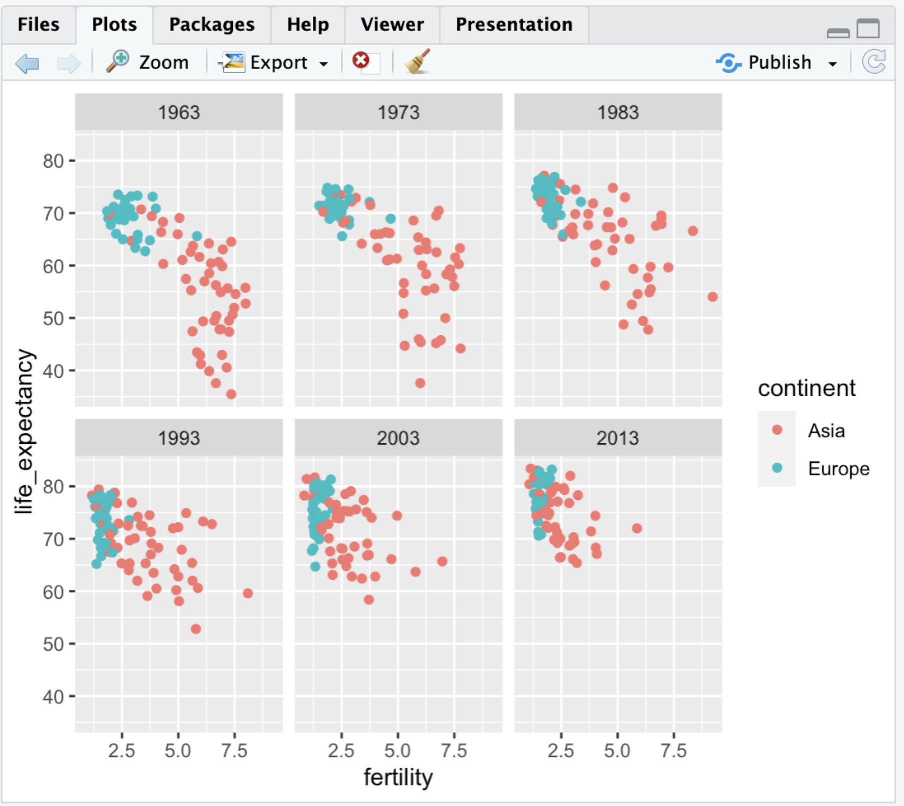
```
> filter(gapminder, year%in%c(1963, 2013)) |>
+ ggplot(aes(fertility, life_expectancy, col = continent)) +
+ geom_point() +
+ facet_grid(. ~ year)
```



This plot clearly shows that the majority of countries have moved from the developing world cluster to the western world one. In 2012, the western versus developing world view no longer makes sense

facet_wrap

To explore how this transformation happened through the years, **we can make the plot for several years**. For example, we can add 1970, 1980, 1990, and 2000. If we do this, we will not want all the plots on the same row, the default behavior of facet_grid, since they will become too thin to show the data. Instead, we will want to use multiple rows and columns. The function facet_wrap permits us to do this by automatically wrapping the series of plots so that each display has viewable dimensions

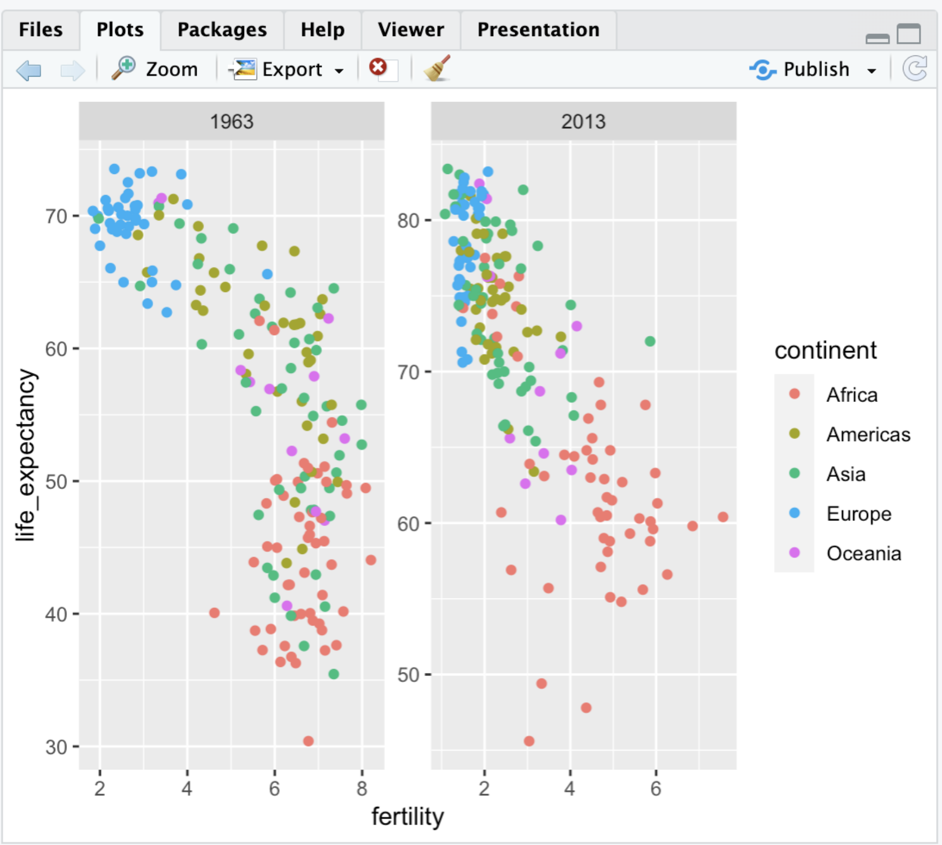



```
R 4.2.2 · ~/
> years <- c(1963, 1973, 1983, 1993, 2003, 2013)
> continents <- c("Europe", "Asia")
> gapminder |>
+ filter(year %in% years & continent %in% continents) |>
+ ggplot(aes(fertility, life_expectancy, col = continent)) +
+ geom_point() +
+ facet_wrap(. ~ year)
```

This plot clearly shows how most Asian countries have improved at a much faster rate than European ones

Fixed scales for better comparisons

The default choice of the range of the axes is important. When not using facet, this range is determined by the data shown in the plot. **When using facet, this range is determined by the data shown in all plots and therefore kept fixed across plots. This makes comparisons across plots much easier.** For example, in the above plot, we can see that life expectancy has increased and the fertility has decreased across most countries. We see this because the cloud of points moves. This is not the case if we adjust the scales



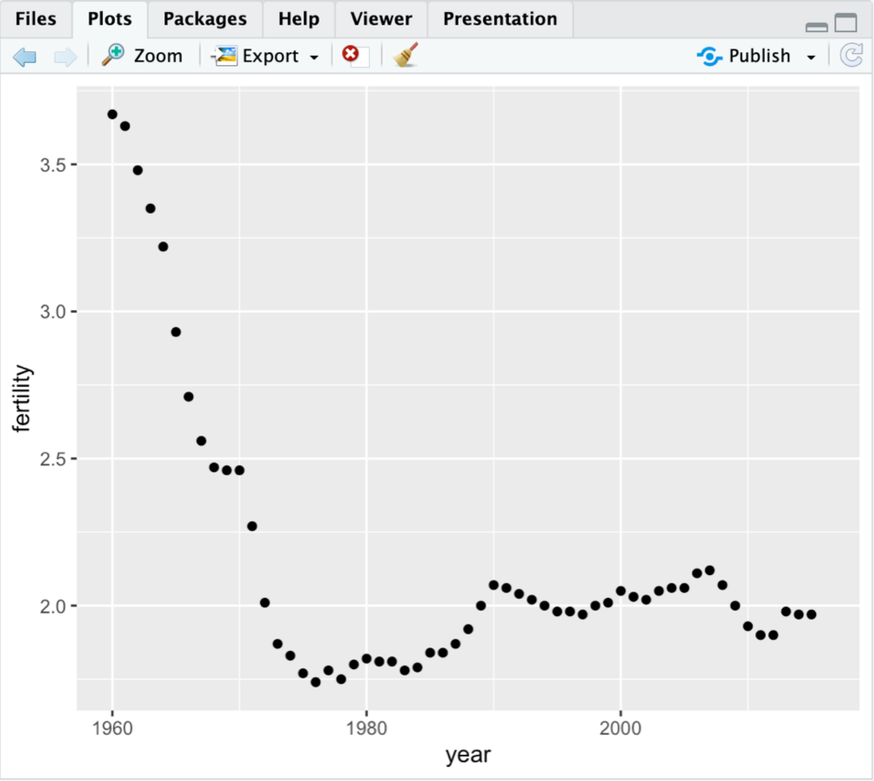
R 4.2.2 · ~/

```
> filter(gapminder, year %in% c(1963, 2013)) |>
+ ggplot(aes(fertility, life_expectancy, col = continent)) +
+ geom_point() +
+ facet_wrap(. ~ year, scales = "free")
```

Time series plots

The visualizations above effectively illustrate that data no longer supports the western versus developing world view. Once we see these plots, new questions emerge. **For example, which countries are improving more and which ones less? Was the improvement constant during the last 50 years or was it more accelerated during certain periods?** For a closer look that may help answer these questions, we introduce time series plots

Time series plots have time in the x-axis and an outcome or measurement of interest on the y-axis. For example, here is a trend plot of United States fertility rates




```
R 4.2.2 · ~/
> gapminder |>
+ filter(country %in% c("United States")) |>
+ ggplot(aes(year, fertility)) +
+ geom_point()
Warning message:
Removed 1 rows containing missing values (`geom_point()`).
```

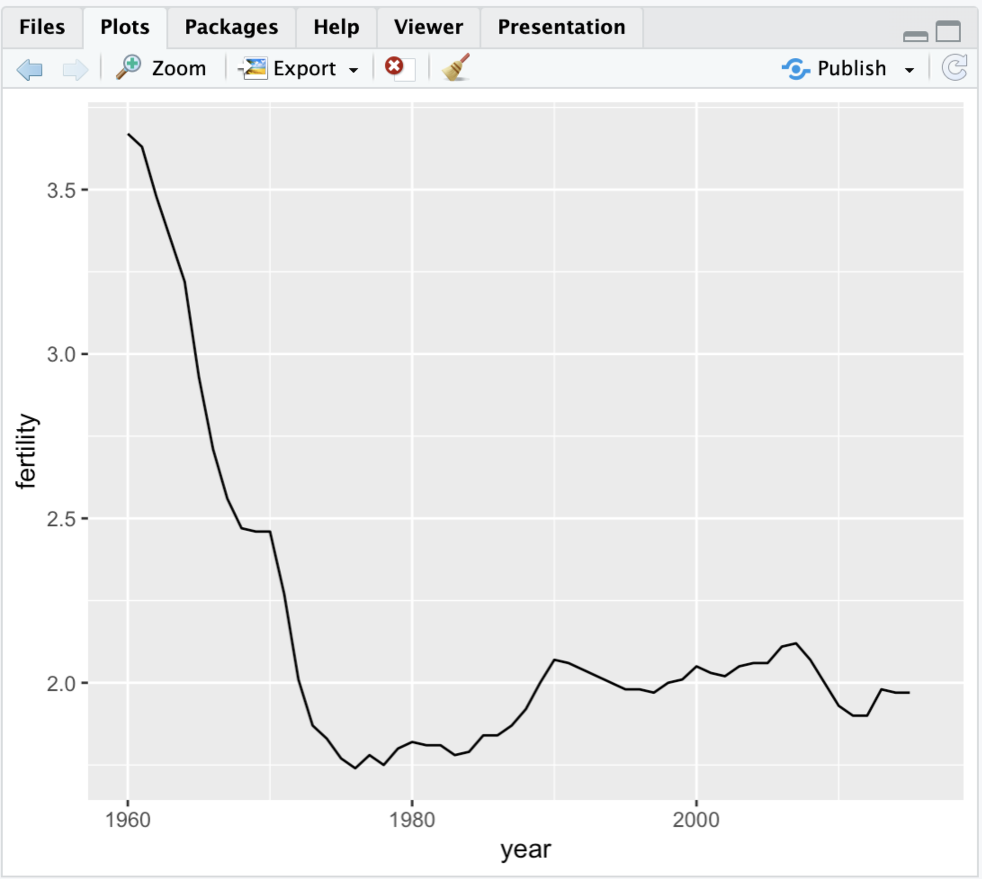
We see that the trend is not linear at all. Instead there is sharp drop during the 60s and 70s to below 2. Then the trend comes back to 2 and stabilizes during the 1990s

Hotmix Gold

La playlist années 70 et 60 – Hotmix Radio

Time series plots

When the points are regularly and densely spaced, we create curves by joining the points with lines, to convey that these data are from a single series, here a country. To do this, we use the geom_line function instead of geom_point



R 4.2.2 · ~/

```
> gapminder |>
+ filter(country %in% c("United States")) |>
+ ggplot(aes(year, fertility)) +
+ geom_line()
Warning message:
Removed 1 row containing missing values (`geom_line()`).
```

Why the warning message?

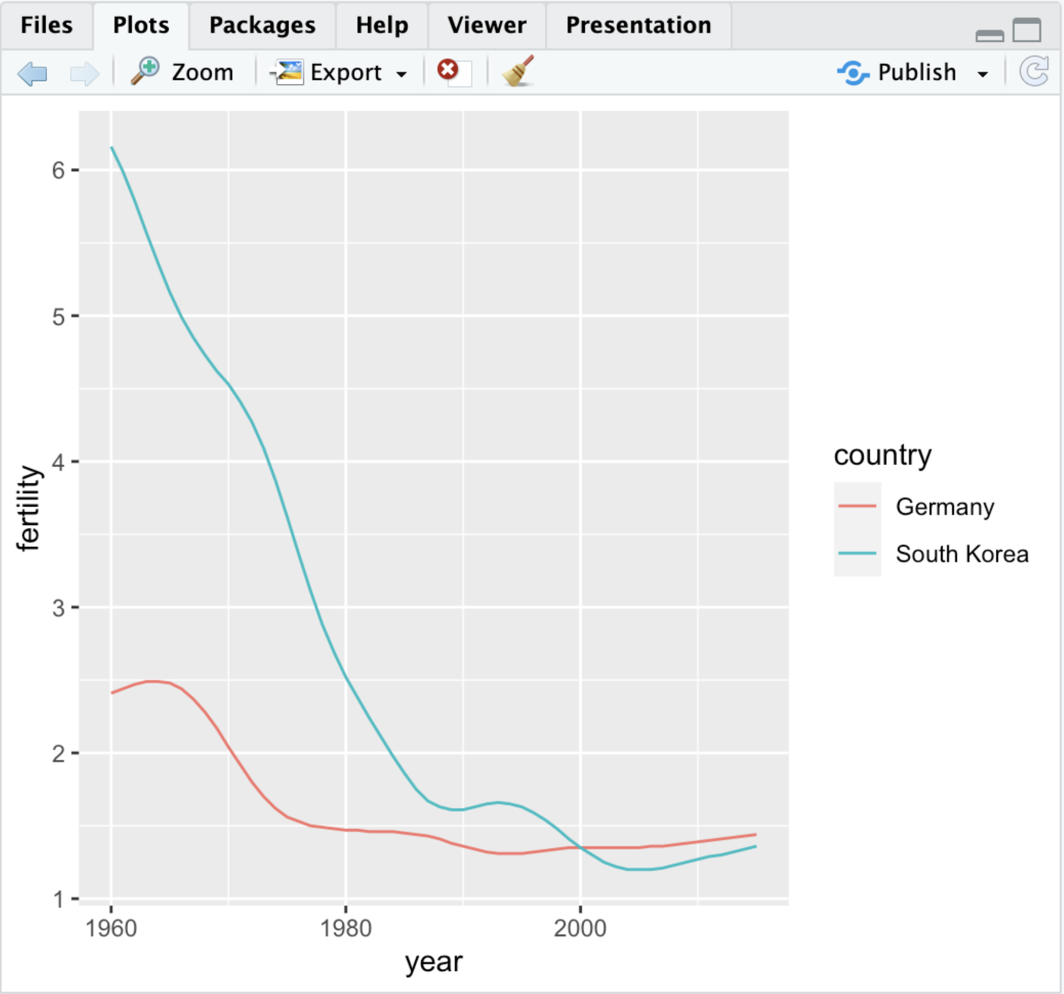
Not all rows are complete!

```
R 4.2.2 · ~/
> filter(gapminder, is.na(gapminder$fertility))
                country year infant_mortality life_expectancy
1             Greenland 2014               NA           72.00
2             Greenland 2015               NA           72.10
3               Albania 2016               NA           78.10
4               Algeria 2016               NA           76.50
5                Angola 2016               NA           60.00
6   Antigua and Barbuda 2016               NA           76.50
7             Argentina 2016               NA           76.70
8               Armenia 2016               NA           74.90
9                 Aruba 2016               NA           75.85
10            Australia 2016               NA           82.30
11              Austria 2016               NA           81.40
12           Azerbaijan 2016               NA           73.30
13              Bahamas 2016               NA           73.90
14              Bahrain 2016               NA           79.10
15           Bangladesh 2016               NA           70.70
16             Barbados 2016               NA           75.80
17              Belarus 2016               NA           71.30
18              Belgium 2016               NA           80.50
19               Belize 2016               NA           71.90
```

Can excluded these rows easily, will learn more ways of handling missing data later in the data wrangling section

Comparing multiple countries

Line plot is particularly helpful when we look at two or more countries, to do that, we need to tell ggplot that we want separate lines.



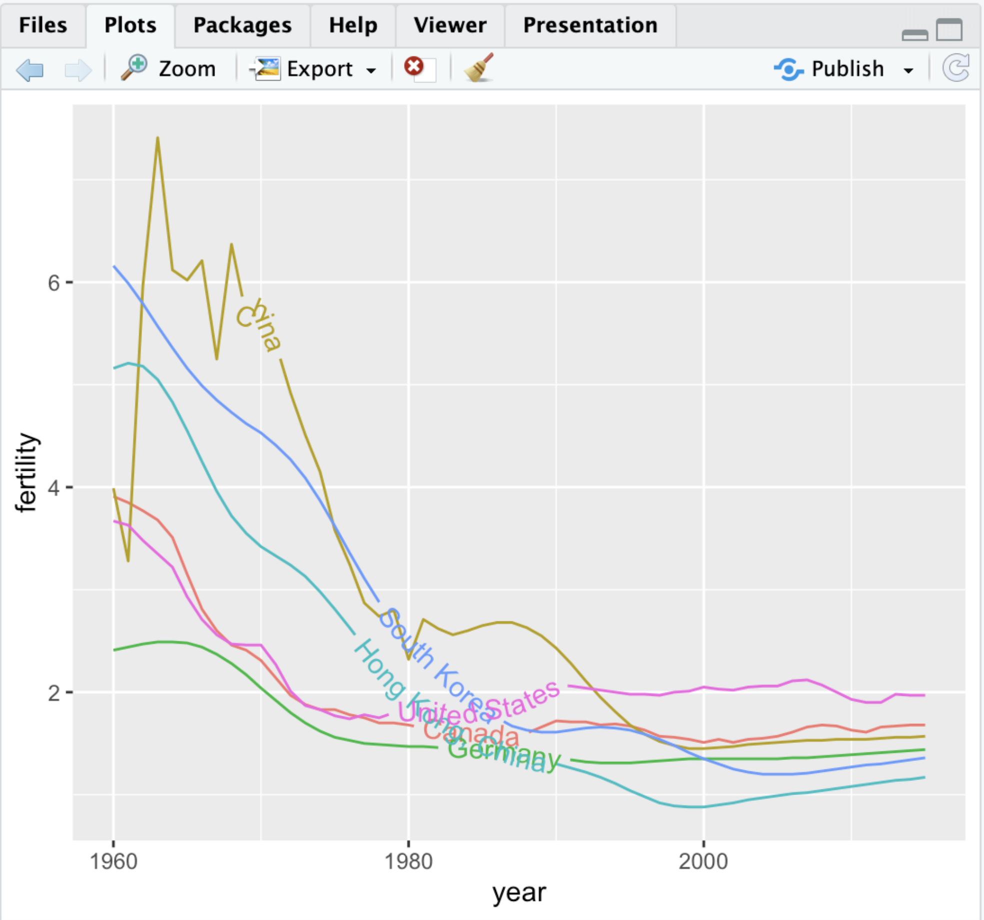
R 4.2.2 · ~/

```
> countries <- c("South Korea","Germany")
> gapminder |>
+ filter(country %in% countries & !is.na(fertility)) |>
+ ggplot(aes(year, fertility, color = country)) +
+ geom_line()
```

The plot clearly shows how South Korea's fertility rate dropped drastically during the 1960s and 1970s, and by 1990 had a similar rate to that of Germany

Labels instead of legends

For trend plots we recommend labeling the lines rather than using legends since the viewer can quickly see which line is which country. This suggestion actually applies to most plots: labeling is usually preferred over legends. **To do that, geomtextpath package is needed**



R 4.2.2 · ~/

```
> gapminder |>
+ filter(country %in% c("United States", "Canada", "Germany", "South K
orea", "China", "Hong Kong, China") & !is.na(fertility)) |>
+ ggplot(aes(year, fertility, color = country, label = country)) +
+ geom_textpath() +
+ theme(legend.position = "none")
```

The gap (fertility) is almost closed between western and non-western countries

Data transformations

We now shift our attention to the second question related to the commonly held notion that wealth distribution across the world has become worse during the last decades:

Has income inequality across countries worsened during the last 40 years?

When general audiences are asked if poor countries have become poorer and rich countries become richer, the majority answers yes. **By using stratification, histograms, smooth densities, and boxplots, we will be able to understand if this is in fact the case.**

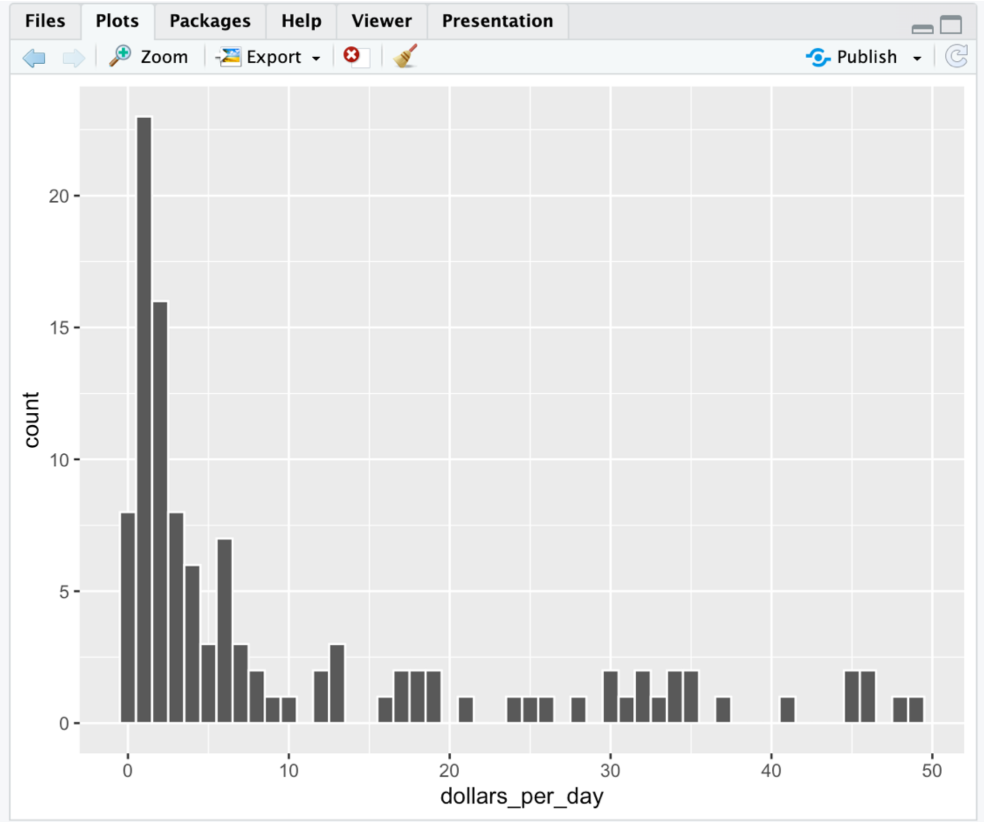
First we learn how transformations can sometimes help provide more informative summaries and plots. The gapminder data table includes a column with the countries' gross domestic product (GDP). GDP measures the market value of goods and services produced by a country in a year. The **GDP per capita** is often used as a rough summary of a country's wealth. Further divided by 365, we have **GDP per capita per day**. Using current US dollars as a unit, a person surviving on **an income of less than $2.15 a day is defined to be living in absolute poverty by the World Bank.**

R 4.2.2 · ~/

```
> gapminder <- gapminder |>  mutate(dollars_per_day = gdp/population/365)
```

With the GDP values in the gapminder dataset adjusted for inflation and all normalized to US dollars, these values are meant to be comparable across the years, but is this transformation good enough?

log transformation

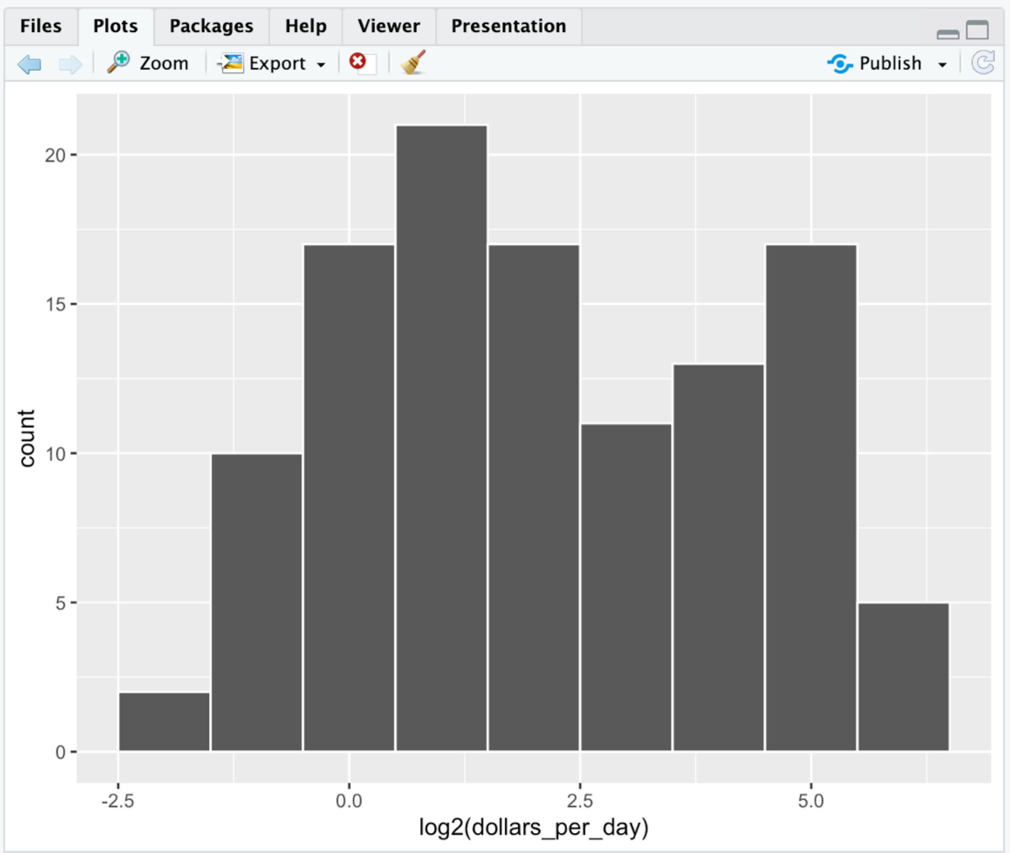


Here is a histogram of per day incomes from 1970

```
R 4.2.2 · ~/
> past_year <- 1970
> gapminder |>
+ filter(year == past_year & !is.na(gdp)) |>
+ ggplot(aes(dollars_per_day)) +
+ geom_histogram(binwidth = 1, color = "white")
```

In this plot, we see that for the majority of countries, averages are below $10 a day. However, the majority of the x-axis is dedicated to the 35 countries with averages above $10. So the plot is not very informative about countries with values below $10 a day

log transformation



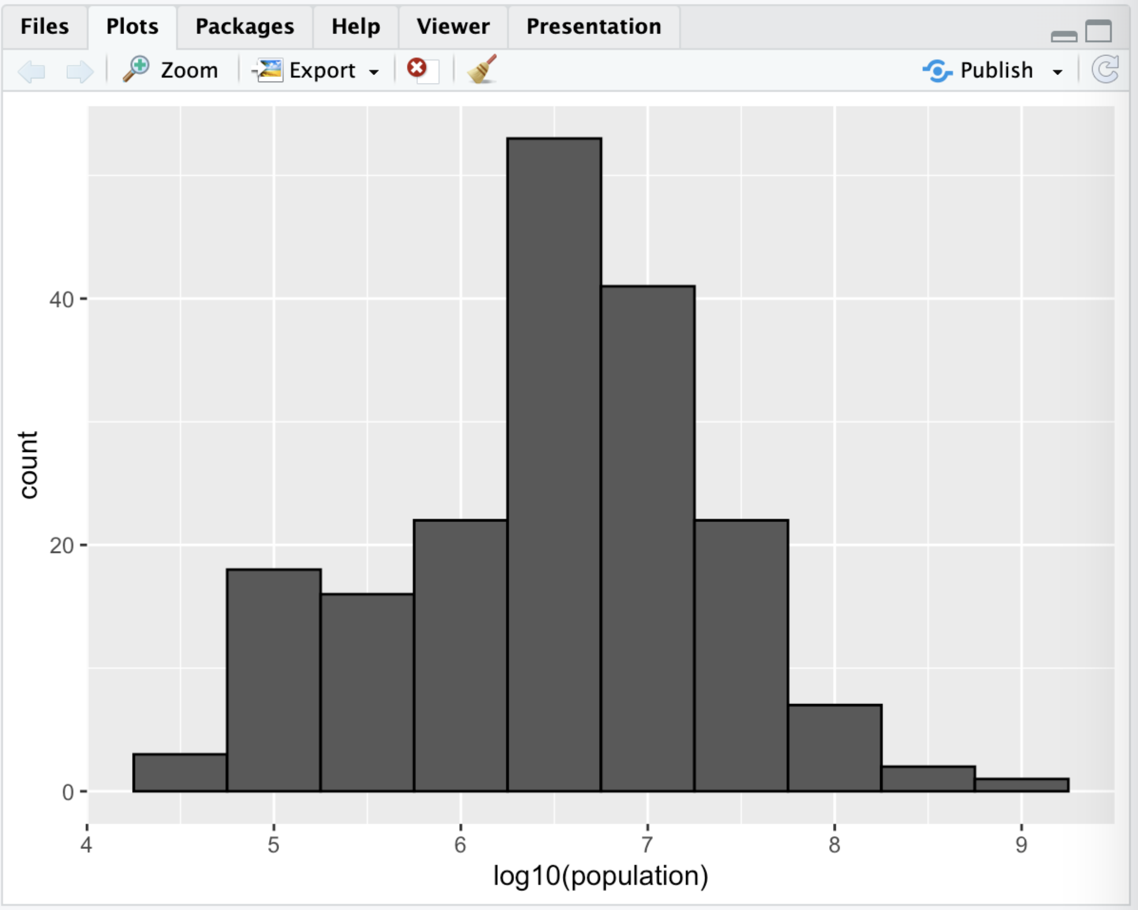
It might be more informative to quickly be able to see how many countries have average daily incomes of about $1 (extremely poor), $2 (very poor), $4 (poor), $8 (middle), $16 (well off), $32 (rich), $64 (very rich) per day. These changes are multiplicative and log transformations convert multiplicative changes into additive ones: when using base 2, a doubling of a value turns into an increase by 1.

Here is the distribution if we apply a log base 2 transform:

```
R 4.2.2 · ~/
> past_year <- 1970
> gapminder |>
+ filter(year == past_year & !is.na(gdp)) |>
+ ggplot(aes(log2(dollars_per_day))) +
+ geom_histogram(binwidth = 1, color = "white")
```

In a way this provides a close-up of the mid to lower income countries

When to use what base?

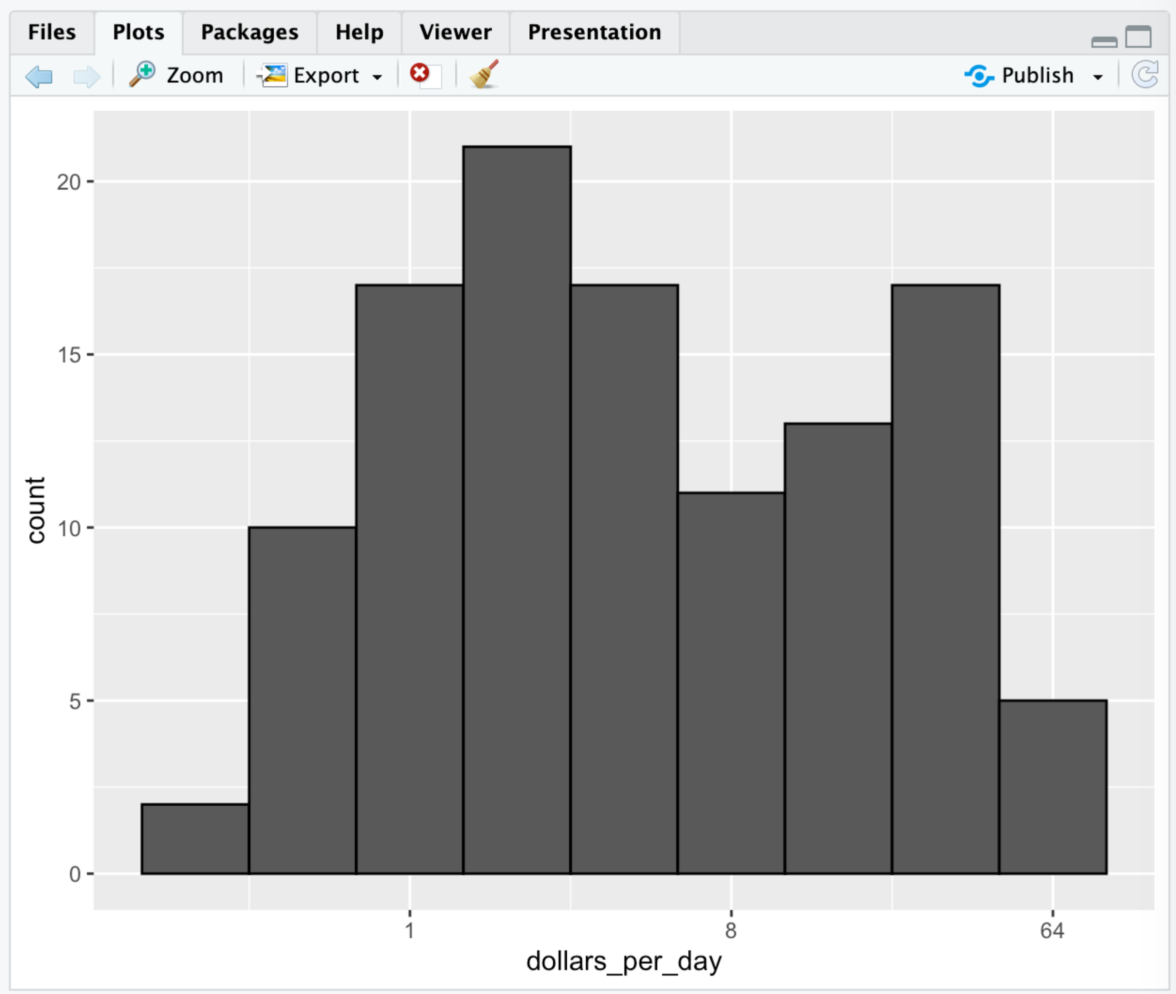


- In the previous case, we used base 2. Other common choices are *e* (the natural log) and 10
- Natural log is not suggested in visualization
- In the example above, the range is [0.327, 48.885], base 2 is better than base 10
- To represent larger values, such as population, base 10 is preferred

```
R 4.2.2 · ~/
> past_year <- 1970
> gapminder |>
+ filter(year == past_year) |>
+ ggplot(aes(log10(population))) +
+ geom_histogram(binwidth = 0.5, color = "black")
```

In the above, we quickly see that country populations range between ten thousand and a billion

Transform the values or the scale



Transform the values:

```
----1----x----2--------3----
```

Transform the scale:

```
----10---x---100------1000---
```

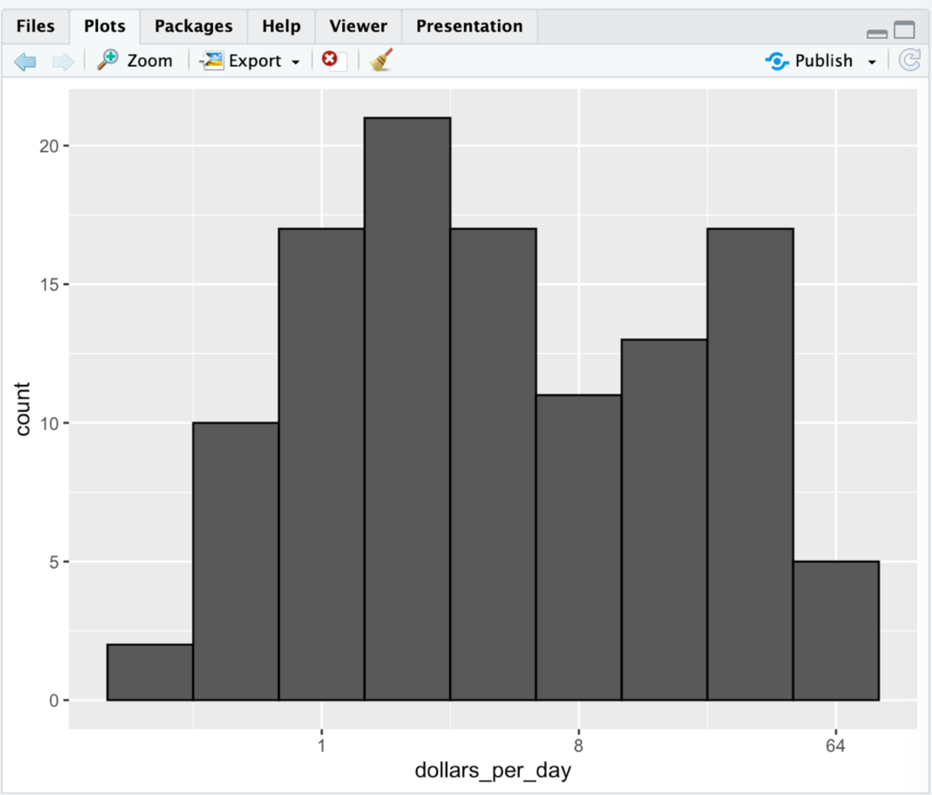
I myself transform the scale most of the time because:

1. The slow speed on transforming values on real large datasets
2. More intuitive axis labels

R 4.2.2 · ~/

```
> gapminder |>
+ filter(year == past_year & !is.na(gdp)) |>
+ ggplot(aes(dollars_per_day)) +
+ geom_histogram(binwidth = 1, color = "black") +
+ scale_x_continuous(trans = "log2")
```

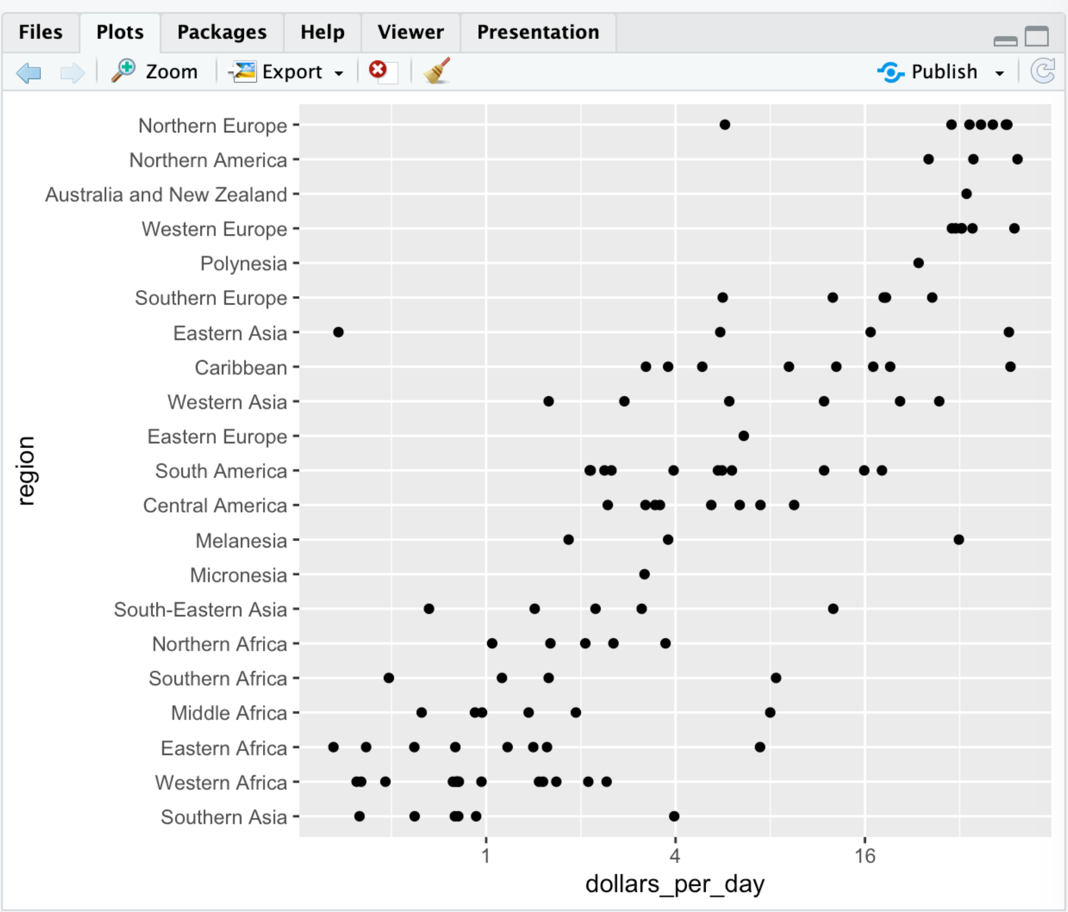
Visualizing multimodal distributions



In the histogram above we see two bumps: one at about 4 and another at about 32. In statistics these bumps are sometimes referred to as **modes**. **The mode of a distribution is the value with the highest frequency**. The mode of the normal distribution is the average. When a distribution, like the one above, doesn't monotonically decrease from the mode, we call the locations where it goes up and down again local modes and say that the distribution has multiple modes

The histogram above suggests that the 1970 country income distribution has two modes: one at about 2 dollars per day (1 in the log 2 scale) and another at about 32 dollars per day (5 in the log 2 scale). This bimodality is consistent with a dichotomous world made up of countries with average incomes less than $8 (3 in the log 2 scale) a day and countries above that

Comparing multiple distributions



A histogram showed us that the 1970 income distribution values show a dichotomy. However, the histogram does not show us if the two groups of countries are west versus the developing world

Let's start by quickly examining the data by region. We reorder the regions by the median value and use a log scale

```
R 4.2.2 · ~/
> gapminder |>
+ filter(year == past_year & !is.na(gdp)) |>
+ mutate(region = reorder(region, dollars_per_day, FUN = median)) |>
+ ggplot(aes(dollars_per_day, region)) +
+ geom_point() +
+ scale_x_continuous(trans = "log2")
```

We can already see that there is indeed a "west versus the rest" dichotomy: we see two clear groups, with the rich group composed of North America, Northern and Western Europe, New Zealand and Australia

Comparing multiple distributions

To facilitate following analyses, we group the countries in to five groups

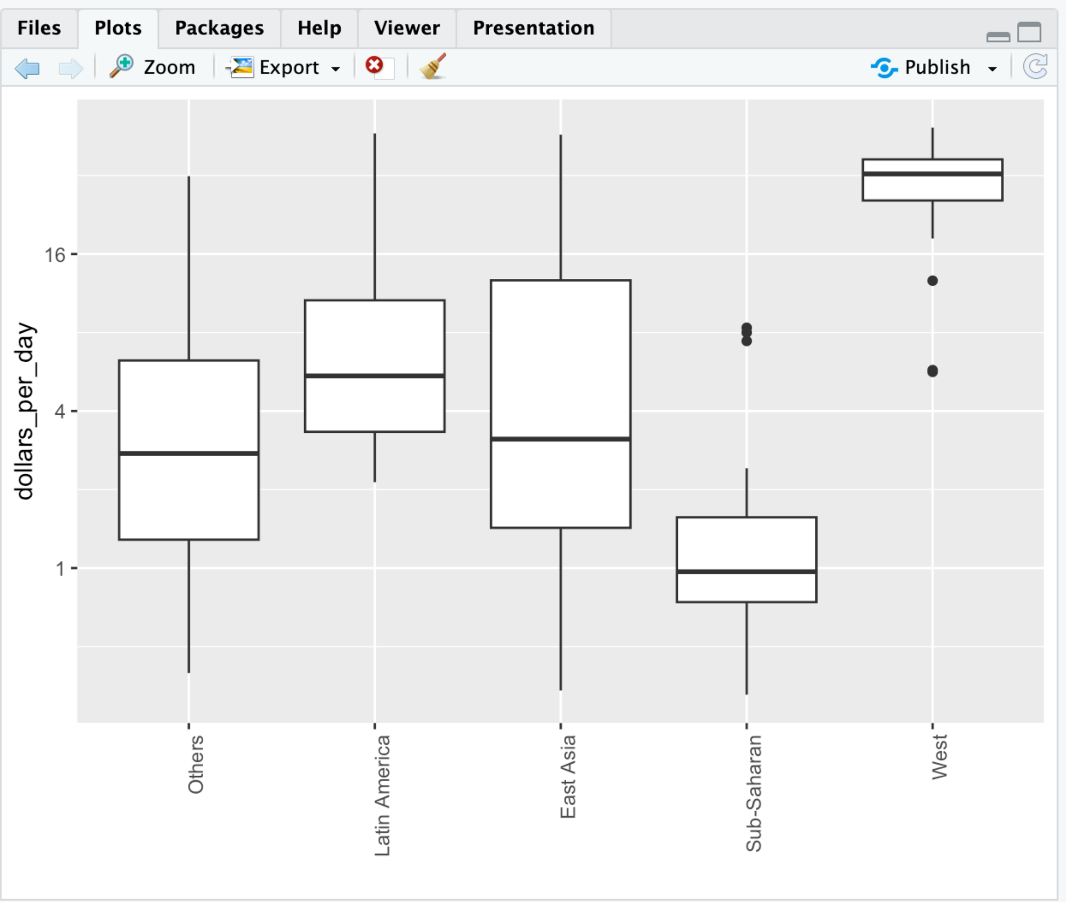
```
R 4.2.2 · ~/
> gapminder <- gapminder |>
+ mutate(group = case_when(
+     region %in% c("Western Europe", "Northern Europe","Southern Europe",
+                   "Northern America", "Australia and New Zealand") ~ "West",
+     region %in% c("Eastern Asia", "South-Eastern Asia") ~ "East Asia",
+     region %in% c("Caribbean", "Central America", "South America") ~ "Latin America",
+     continent == "Africa" & region != "Northern Africa" ~ "Sub-Saharan",
+     TRUE ~ "Others")
+ )
```

We turn this group variable into a factor to control the order of the levels

```
R 4.2.2 · ~/
> gapminder <- gapminder |>
+ mutate(group = factor(group,
+                       levels = c("Others", "Latin America",
+                                  "East Asia", "Sub-Saharan",
+                                  "West")))
+
```

```
R 4.2.2 · ~/
> class(gapminder$group)
[1] "factor"
```

Comparing multiple distributions through boxplot

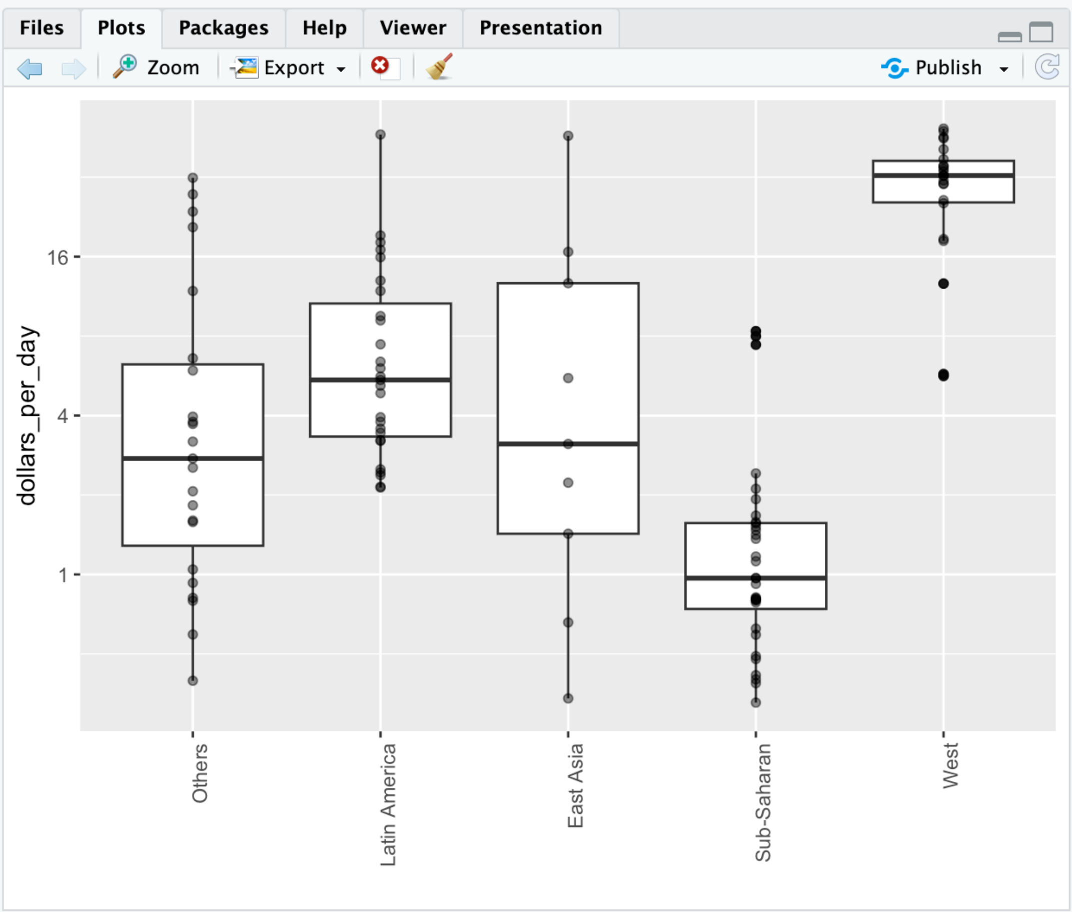


R 4.2.2 · ~/

```
> p <- gapminder |>
+     filter(year == past_year & !is.na(gdp)) |>
+     ggplot(aes(group, dollars_per_day)) +
+     geom_boxplot() +
+     scale_y_continuous(trans = "log2") +
+     xlab("") +
+     theme(axis.text.x = element_text(angle = 90, hjust = 1))
> p
```

Right-aligned

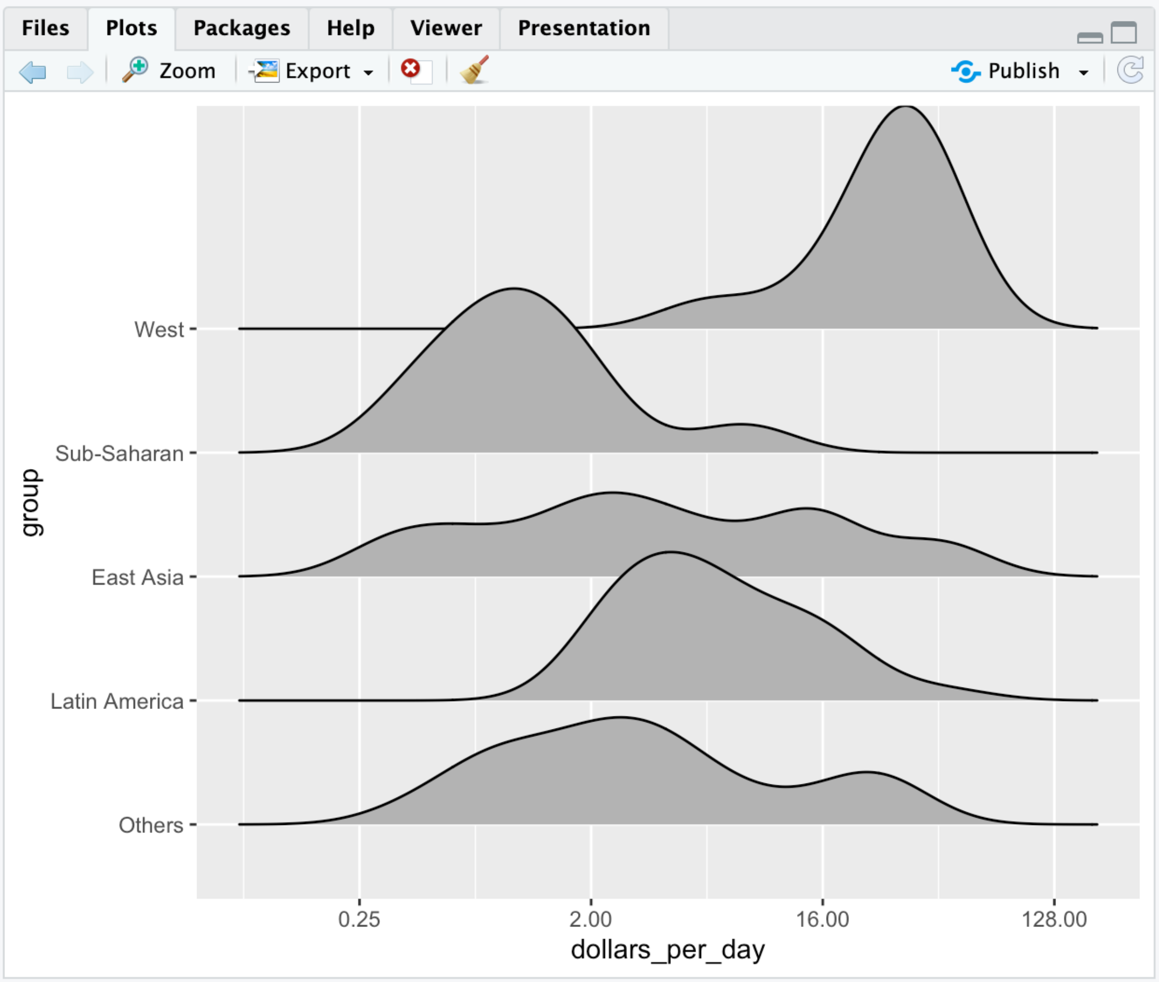
Comparing multiple distributions through boxplot



Boxplots have the limitation that by summarizing the data into five numbers, we might miss important characteristics of the data. One way to avoid this is by showing the data

This is also called swarm plot

Comparing multiple distributions through ridge plot



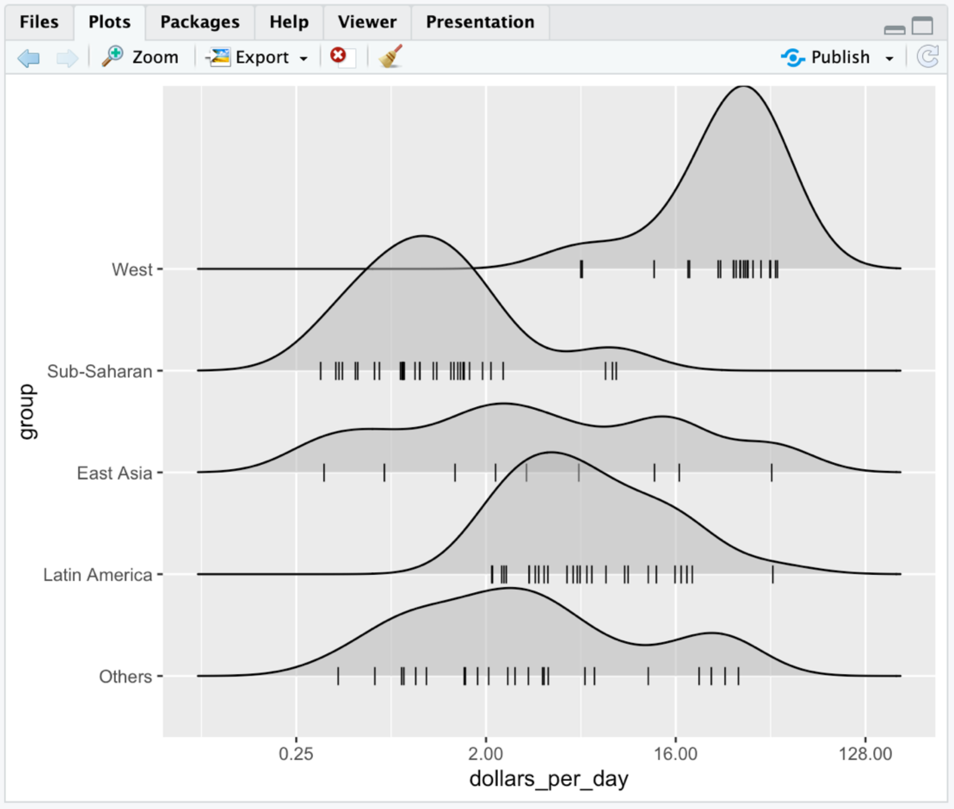
- Ridge plot is stacked smooth densities
- Package ggridges is needed

R 4.2.2 · ~/

```
> library(ggridges)
> p <- gapminder |>
+      filter(year == past_year & !is.na(dollars_per_day)) |>
+      ggplot(aes(dollars_per_day, group)) +
+      scale_x_continuous(trans = "log2")
> p  + geom_density_ridges()
Picking joint bandwidth of 0.648
```

Interval adjustable through "scale ="

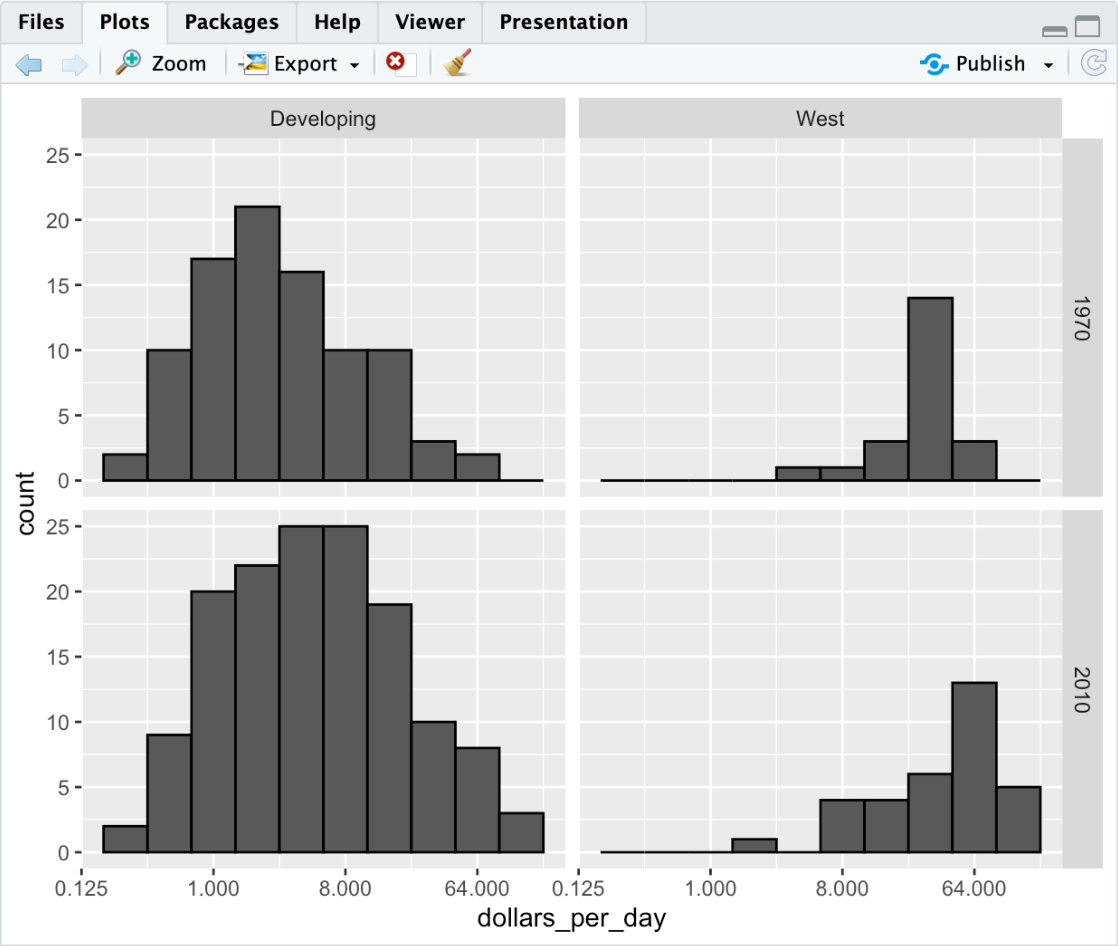
Ridge plot with rug representation



R 4.2.2 · ~/

```
> p <- gapminder |>
+     filter(year == past_year & !is.na(dollars_per_day)) |>
+     ggplot(aes(dollars_per_day, group)) +
+     scale_x_continuous(trans = "log2")
> p  + geom_density_ridges(jittered_points = TRUE,
+                          position = position_points_jitter(height = 0),
+                          point_shape = '|', point_size = 3,
+                          point_alpha = 1, alpha = 0.5)
Picking joint bandwidth of 0.648
```

1970 versus 2010 income distributions

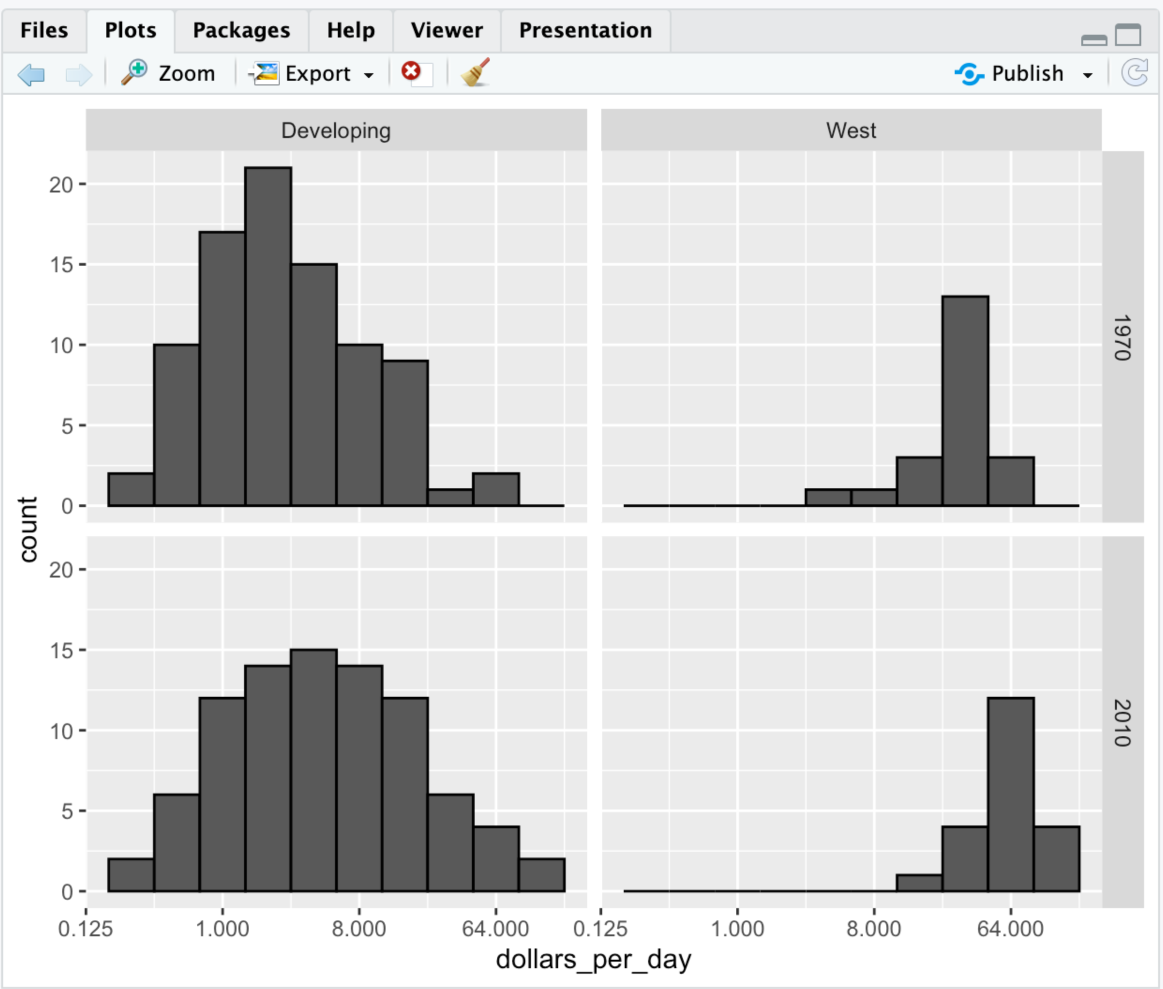


Data exploration clearly shows that in 1970 there was a "west versus the rest" dichotomy. But does this dichotomy persist? Let's use facet_grid to see how the distributions have changed. To start, we will focus on two groups: the west and the rest. We make four histograms

```
R 4.2.2 · ~/
> past_year <- 1970
> present_year <- 2010
> years <- c(past_year, present_year)
> gapminder |>
+ filter(year %in% years & !is.na(gdp)) |>
+ mutate(west = ifelse(group == "West", "West", "Developing")) |>
+ ggplot(aes(dollars_per_day)) +
+ geom_histogram(binwidth = 1, color = "black") +
+ scale_x_continuous(trans = "log2") +
+ facet_grid(year ~ west)
```

The plot seems to have more countries in 2010 than 1970?

1970 versus 2010 income distributions



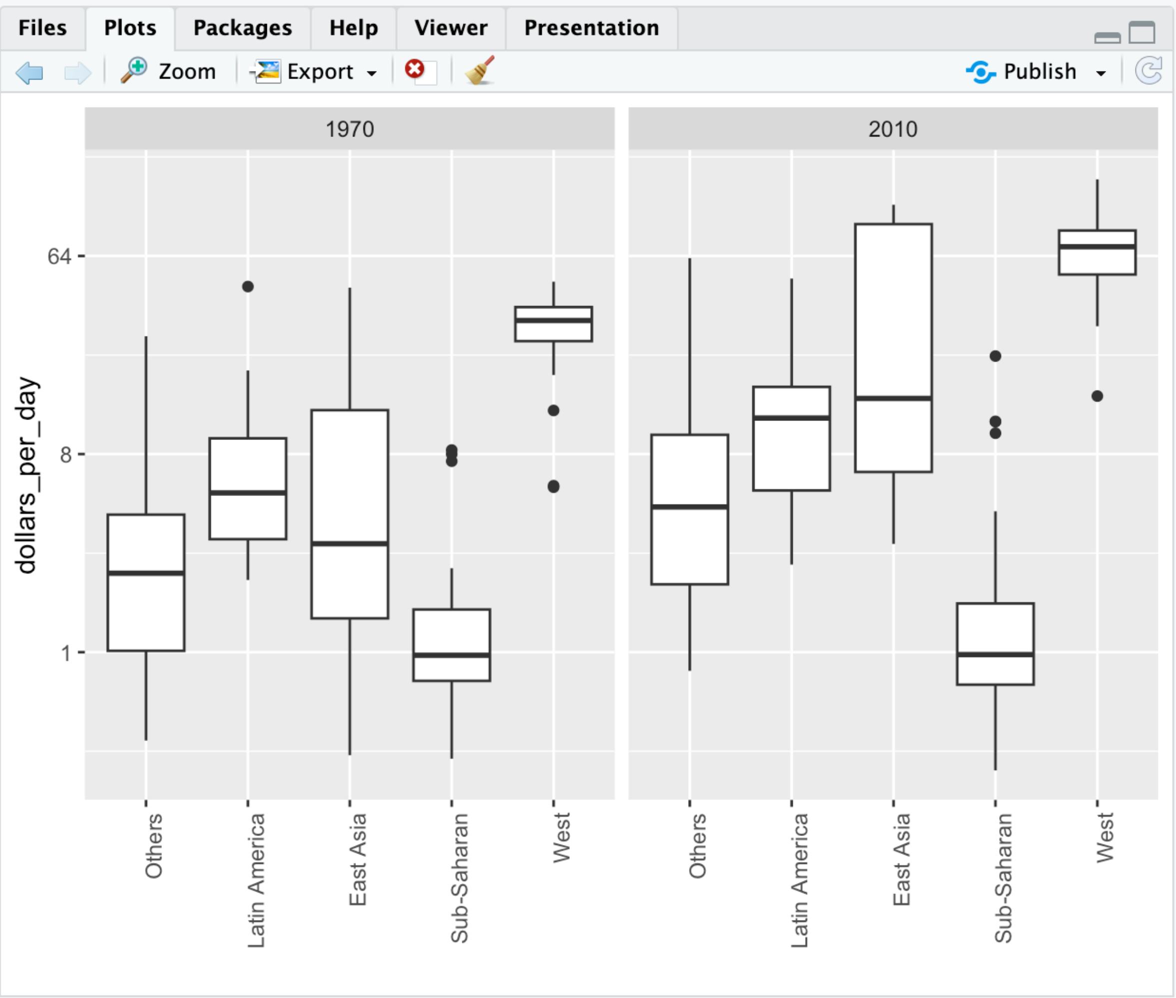
We remake the plots using only countries with data available for both years. In the data wrangling part of this book, we will learn tidyverse tools that permit us to write efficient code for this, but here we can use simple code using the intersect function

```
R 4.2.2 · ~/
> country_list_1 <- gapminder |>
+ filter(year == past_year & !is.na(dollars_per_day)) |>
+ pull(country)
>
> country_list_2 <- gapminder |>
+ filter(year == present_year & !is.na(dollars_per_day)) |>
+ pull(country)
>
> country_list <- intersect(country_list_1, country_list_2)
```

```
R 4.2.2 · ~/
> length(country_list_1)
[1] 113
> length(country_list_2)
[1] 176
> length(country_list)
[1] 108
```

```
filter(year %in% years & !is.na(gdp) & country %in% country_list) |>
```

1970 versus 2010 income distributions



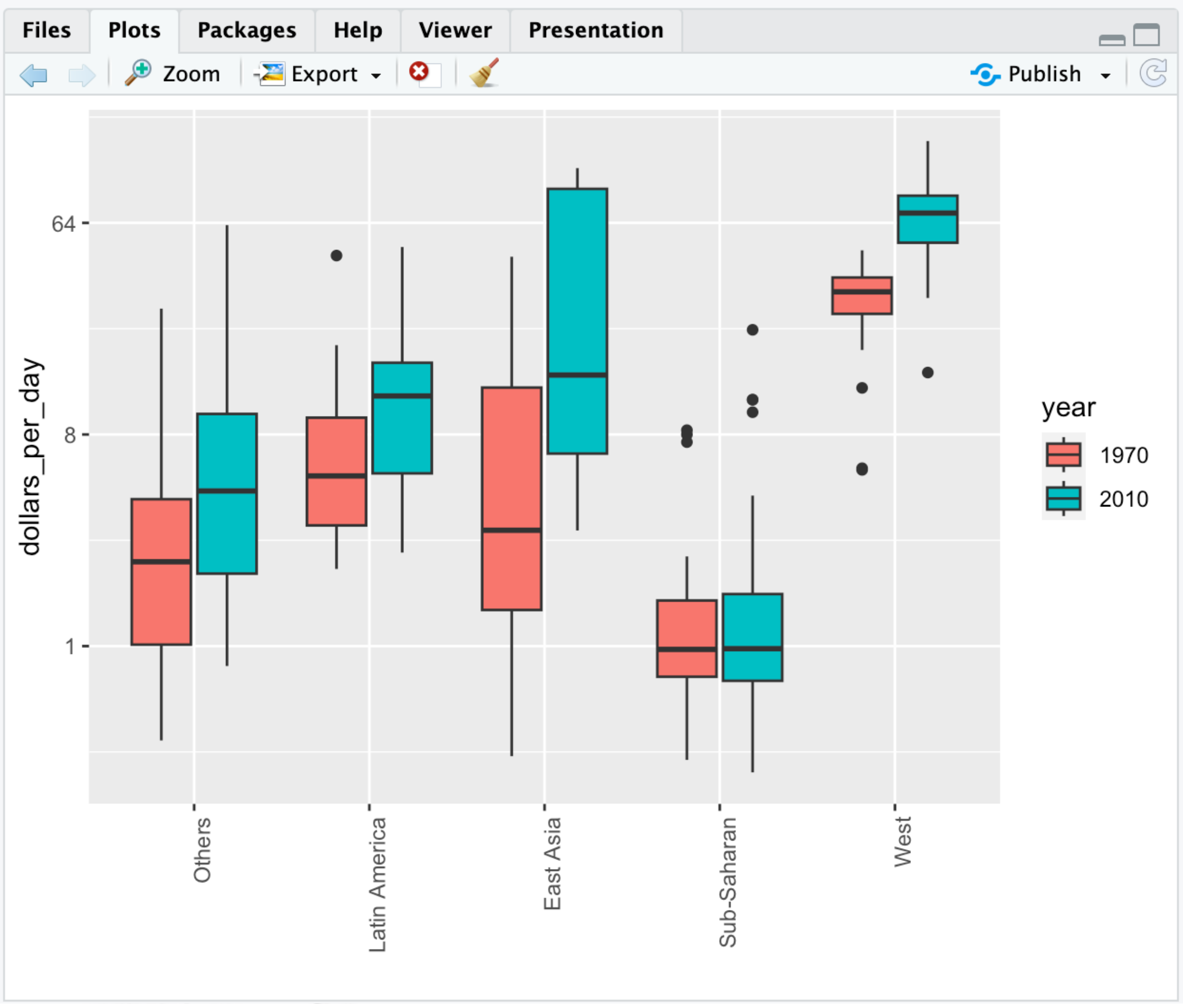
We now see that the rich countries have become a bit richer, but percentage-wise, the poor countries appear to have improved more

To see which specific regions improved the most, we can remake the boxplots we made above, but now adding the year 2010 and then using facet to compare the two years

R 4.2.2 · ~/

```
> gapminder |>
+ filter(year %in% years & country %in% country_list) |>
+ ggplot(aes(group, dollars_per_day)) +
+ geom_boxplot() +
+ theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
+ scale_y_continuous(trans = "log2") +
+ xlab("") +
+ facet_grid(. ~ year)
```

1970 versus 2010 income distributions

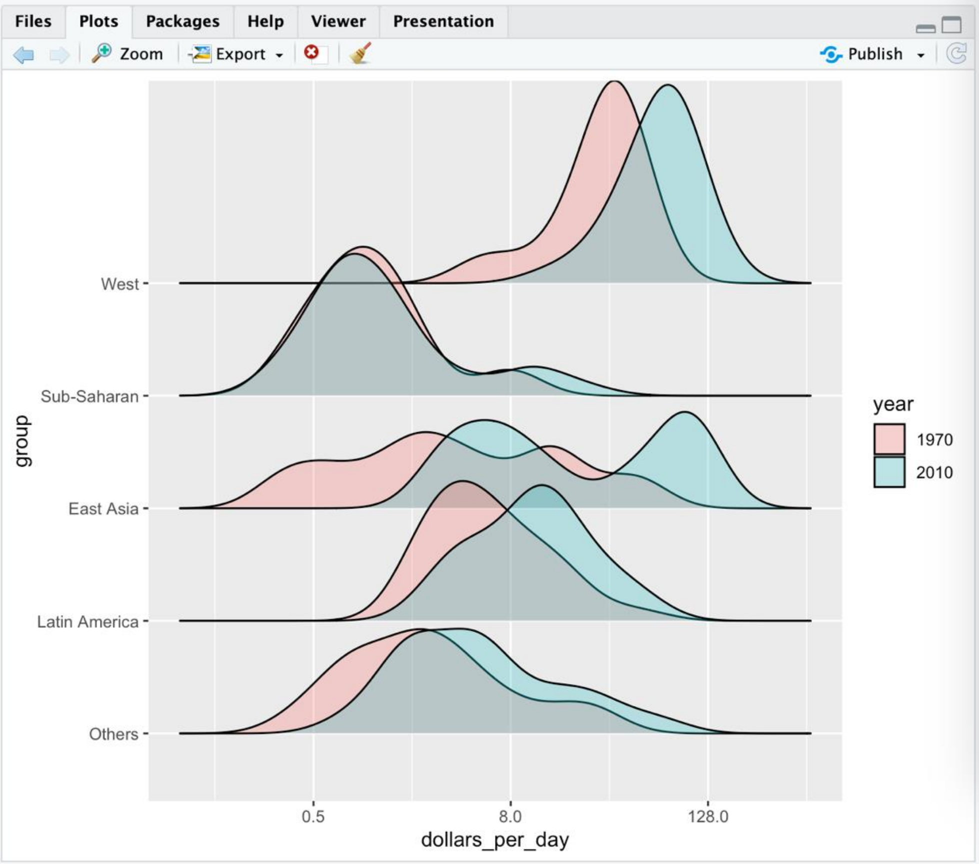


More conveniently, we can compare the two years at each region side by side, this requires using aes(fill=year). However, fill= accepts only factor, thus, we convert year into factor after filter

R 4.2.2 · ~/

```
> gapminder |>
+ filter(year %in% years & country %in% country_list) |>
+ mutate(year = factor(year)) |>
+ ggplot(aes(group, dollars_per_day, fill = year)) +
+ geom_boxplot() +
+ theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
+ scale_y_continuous(trans = "log2") +
+ xlab("")
```

1970 versus 2010 income distributions

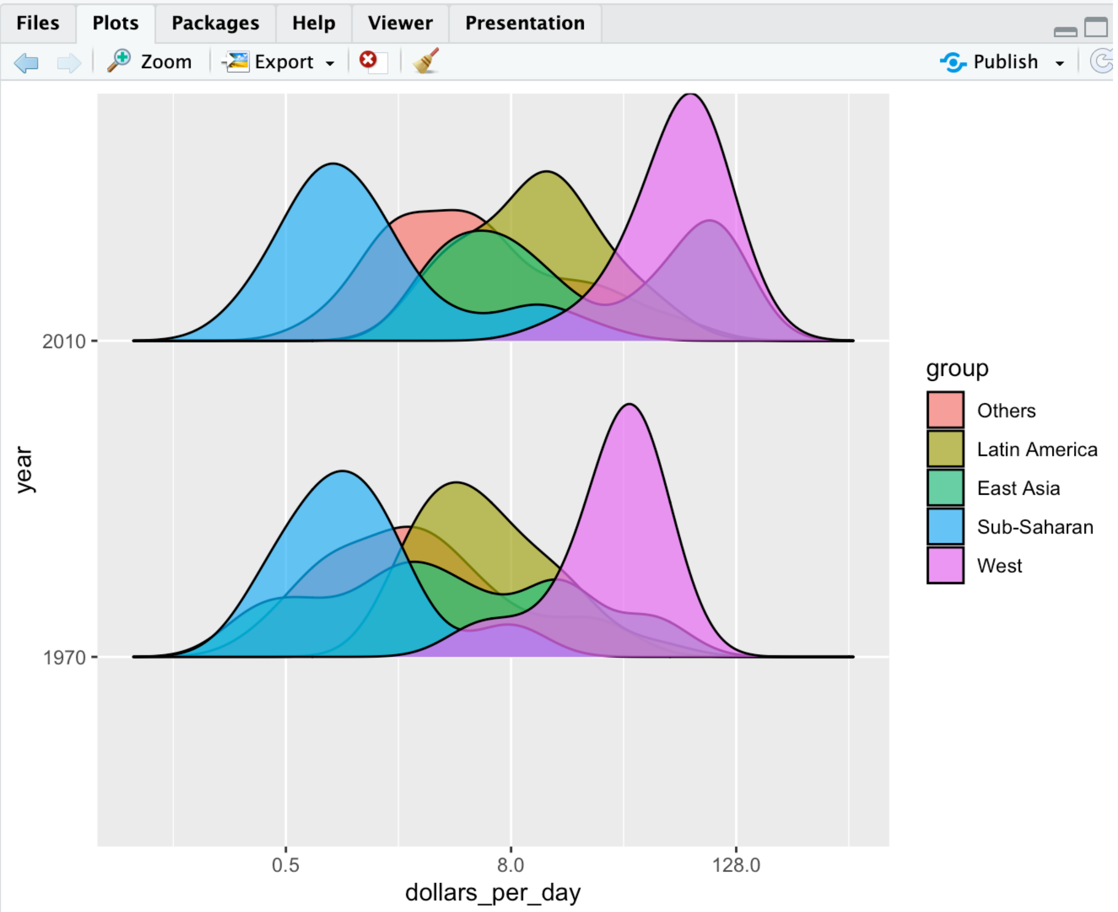


R 4.2.2 · ~/

```
> p <- gapminder |>
+      filter(year %in% years & !is.na(dollars_per_day) &
+             country %in% country_list) |>
+      mutate(year = factor(year)) |>
+      ggplot(aes(dollars_per_day, group, fill=year)) +
+      scale_x_continuous(trans = "log2")
> p  + geom_density_ridges(alpha = 0.3)
Picking joint bandwidth of 0.639
```

Using ridge plot on multiple regions and years, we found that East Asia has improved most significantly, but still seems to have to modes

1970 versus 2010 income distributions



Switching year and region for a ridge plot, seems not a good move, but what if we have more than 5 years to compare between?

R 4.2.2 · ~/

```
> p <- gapminder |>
+     filter(year %in% years & !is.na(dollars_per_day) &
+            country %in% country_list) |>
+     mutate(year = factor(year)) |>
+     ggplot(aes(dollars_per_day, year, fill=group)) +
+     scale_x_continuous(trans = "log2")
> p  + geom_density_ridges(alpha = 0.7, scale = 0.8)
Picking joint bandwidth of 0.639
```